

Investigating Interfacial Mechanisms and Switching Behavior in $\text{Ge}_1\text{Sb}_2\text{Te}_4$ / Sb_2Te_3 Superlattices

by

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MOTIVATION

The continuous advancement of technology and the exponential growth of digitally stored information have significantly increased the demand for energy-efficient, high-capacity data storage solutions. [1] Traditional von-Neumann computer architectures, which separate memory and computation units, face fundamental limitations due to the data transfer bottleneck between processing and memory units. This challenge, coupled with the plateauing performance gains from classical scaling described by Moore's Law, has driven the search for alternative memory concepts that offer higher performance and efficiency.

One promising approach to overcome these limitations is the concept of in-memory computing and storage-class memory (SCM). SCM aims to bridge the performance gap between volatile dynamic random-access memory (DRAM) and non-volatile Flash memory by providing fast, non-volatile storage with high endurance (Figure 1.1). Among the most promising materials enabling SCM are Phase-Change Materials (PCMs), which can reversibly switch between amorphous and crystalline states, offering distinct electrical resistivities that can encode data.

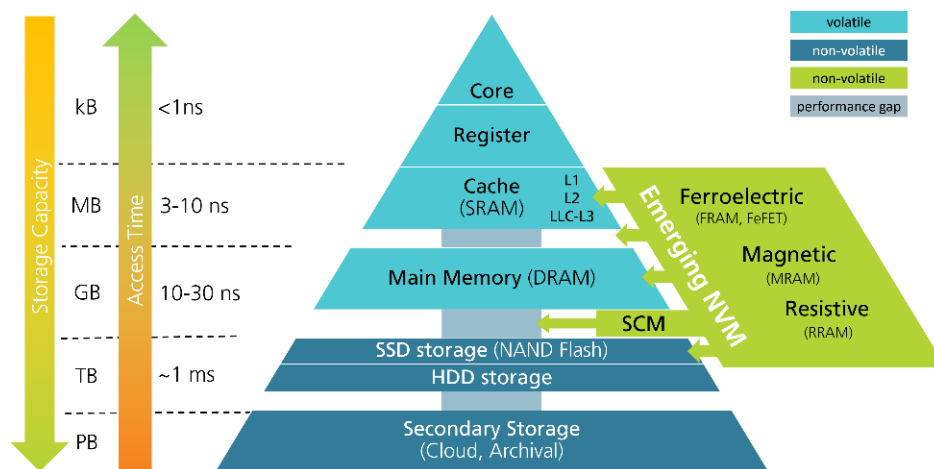


Figure 1.1: Memory hierarchy: DRAM and flash with their opposing characteristics that leave a gap to be filled by storage-class memory [2]

PCMs are also a leading material candidate for neuromorphic applications due to their ability to simultaneously process and store data, making them highly suitable for brain-inspired computing. As neuromorphic systems aim to replicate the functionality of biological neurons and synapses, PCMs stand out for their unique physical properties, including threshold switching and progressive crystallization/amorphization. These characteristics enable PCMs to mimic synaptic plasticity and neuronal dynamics, essential for efficient and energy-saving computation. [3]

PCMs based on chalcogenide materials such as $\text{Ge}_2\text{Sb}_2\text{Te}_5$ have been widely studied as promising candidates for next-generation data storage technologies due to their non-volatility, fast switching speeds, and high cyclability. Despite their advantages, conventional PCM devices suffer from high programming currents and limited endurance. [4]

A breakthrough in PCM technology was achieved by Simpson et al., who demonstrated that epitaxially matched superlattices of $\text{GeTe}/\text{Sb}_2\text{Te}_3$ enable ~ 3 times faster switching speeds, an order of magnitude improved endurance, and an order of magnitude reduction in power consumption compared to conventional PCMs.[5] These PCM superlattices are hypothesized to switch through a novel mechanism driven by local atomic rearrangements at the interfaces, rather than melt-quenching, though this crystalline-crystalline switching has yet to be conclusively proven. [5, 6]

Recent studies have further expanded the scope of superlattice-based PCMs by achieving up to a $10\times$ reduction in RESET current density in $\text{GeTe}/\text{Sb}_2\text{Te}_3$ systems, a significant improvement over conventional GST-based PCMs. Kahn et al. attribute this improvement to electro-thermal confinement effects caused by van der Waals-like interfaces, revisiting the melt-quench mechanism hypothesis. They further emphasize the potential of these superlattices for low-power, high-density memory and neuromorphic computing applications. [7] However, despite the rapid progress in this field, the exact switching mechanisms and interfacial dynamics in chalcogenide superlattices remain under intense scientific debate.

This thesis aims to contribute to the ongoing exploration of superlattices by investigating the switching behavior of highly textured, MBE-grown $\text{Ge}_1\text{Sb}_2\text{Te}_4/\text{Sb}_2\text{Te}_3$ superlattices using a pulsed laser system. This optical switching of PCMs is particularly intriguing for neuromorphic photonics, an interdisciplinary research field that explores complex photonic devices that have the functionality of optical neuromorphic computing and has yet to be explored for superlattices. Optical systems offer advantages over electronic systems, such as higher speed, lower power consumption, and reduced crosstalk, making it a promising approach for developing advanced photonic devices with neuromorphic computing capabilities. [8] Atom Probe Tomography (APT) is used to compare the structural and chemical characteristics of interfaces in recrystallized and as-deposited regions. APT's three-dimensional, near-atomic resolution reveals changes in morphology and element

distribution. The insights gained from this study will help optimize PCM devices, enhancing their performance, efficiency, and reliability for future memory applications.

THEORETICAL CONCEPTS OF PCM SUPERLATTICES

2.1 Phase Change Materials (PCMs)

Phase-change materials (PCMs) are a class of chalcogenides that can reversibly switch between amorphous and crystalline states, making them highly valuable for data storage and computing applications. This transformation is accompanied by a significant contrast in electrical and optical properties, with the crystalline phase exhibiting at least two orders of magnitude higher conductivity and 15% reflectance contrast compared to its amorphous counterpart. [9, 10] These notable features stem from the distinct structural and bonding properties. Moreover, PCMs enable non-volatile switching, meaning they can retain their state without a constant power supply, ensuring long-term data stability under ambient conditions. Their exceptionally fast phase transition kinetics, with crystallization occurring within nanoseconds at elevated temperatures, further enhances their suitability for high-speed memory applications and next-generation computing technologies.

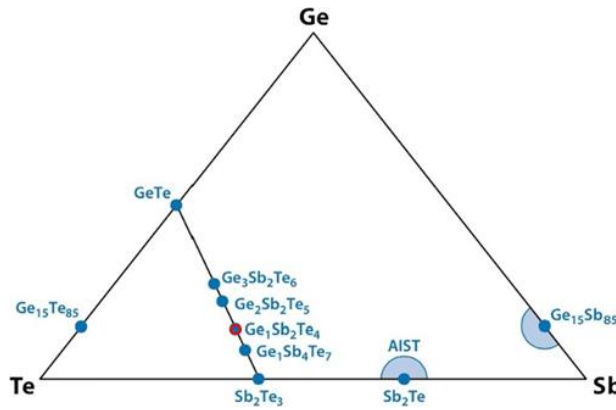


Figure 2.1: The ternary phase diagram of the Ge-Sb-Te system. the diagram highlights the pseudo-binary line between GeTe and Sb_2Te_3 , where key phase-change materials such as $\text{Ge}_2\text{Sb}_2\text{Te}_5$ (GST-225) and $\text{Ge}_1\text{Sb}_2\text{Te}_4$ (GST-124), which will be investigated in this thesis, are located. [11]

PCMs switching behavior gained significant attention with the use of chalcogenide materials along the pseudo-binary line between GeTe and Sb_2Te_3 , with $\text{Ge}_2\text{Sb}_2\text{Te}_5$ (GST)

becoming one of the most studied and utilized compounds for phase-change storage technology. As shown in the ternary phase diagram of the Ge-Sb-Te system (Figure 2.1), several stable phases of Germanium-Antimony-Telluride compounds exist along this pseudo-binary line, with compositions of $(\text{GeTe})_m(\text{Sb}_2\text{Te}_3)_n$.

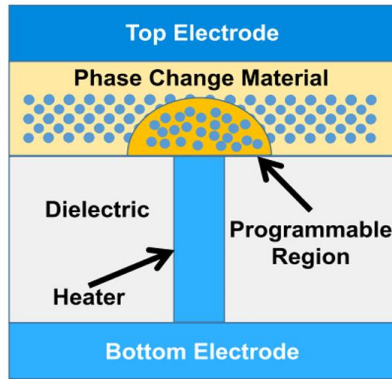


Figure 2.2: Schematic representation of a PC mushroom cell. During the RESET operation, an amorphous mark is melt-quenched above the heater defining the RESET state. [12]

Phase-change materials (PCMs) can transition between amorphous and crystalline states by heating the system and applying electrical or optical pulses. This electrically induced transition is typically implemented in a mushroom cell geometry, where a small active area atop a narrow bottom electrode concentrates heat, enhancing energy efficiency and enabling localized, rapid phase changes. (Figure 2.2) The switching mechanism relies on Joule heating. In the RESET operation, a high-current pulse melts the material, and rapid quenching traps it in a high-resistance, amorphous state.

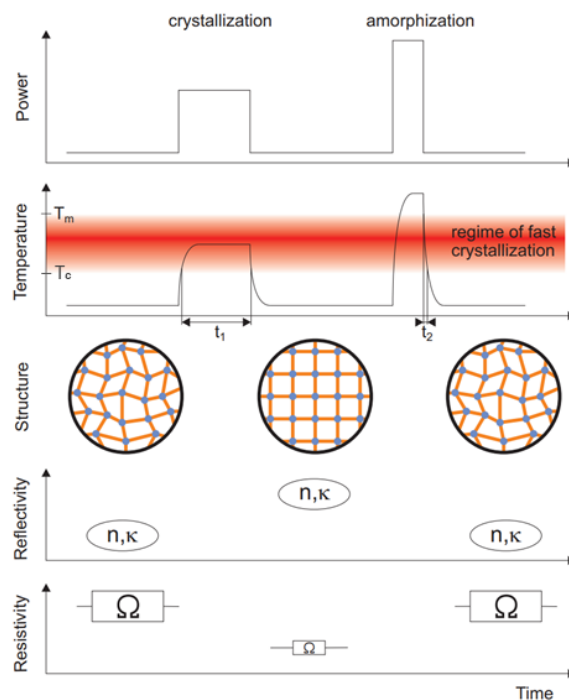


Figure 2.3: Switching principle of PCMs. When heated above the crystallization temperature (T_c), the material crystallizes within a time t_1 , dictated by its temperature-dependent growth velocity. Amorphization occurs by melting ($T > T_m$) and rapidly cooling (minimizing t_2) to bypass crystallization through melt-quenching. [13]

The SET operation applies a lower-energy pulse above the glass transition temperature, allowing the material to recrystallize into a low-resistance state. This transition is enabled by threshold switching, where the amorphous phase experiences a sudden drop in resistance when a critical voltage (V_t) is reached. The flow of this operation, including the applied power pulses, resulting temperature pulse, structural changes, and corresponding variations in reflectivity and resistivity, is depicted in Figure 2.3.

The differences in the two phases can be traced back to their different bonding mechanisms. In the crystalline phase, PCMs exhibit a unique bonding mechanism known as metavalent bonding, which presents different properties from other bonding mechanisms. This mechanism involves delocalized p-electrons forming bonds with multiple neighbors, resulting in high polarizability and a pseudo-cubic unit cell. However, this bonding requires long-range order, which is lost in the amorphous phase, where atoms instead form ordinary covalent bonds, satisfying the 8-N rule. [10, 14] The distinction between metavalent and covalent bonding is further illustrated by quantum chemical bond descriptors, which classify PCMs within a distinct region of a bonding map based on electron transfer (ET) and electron sharing (ES). (Figure 2.4) [14] This insight not only aids in predicting optimal phase-change materials but also underscores the fundamental role of bonding in their properties.

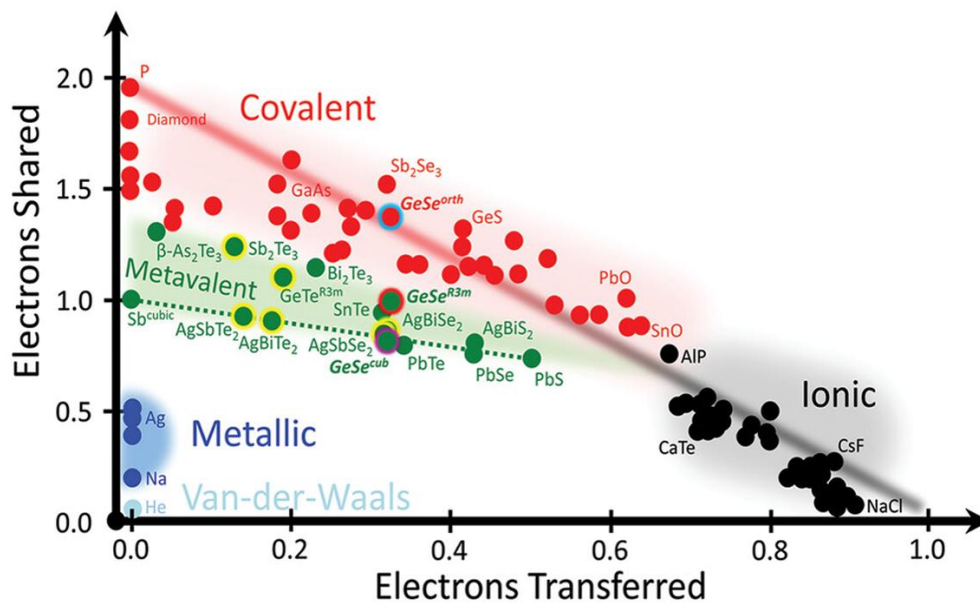


Figure 2.4: Bonding map using the quantum chemical bond descriptors ES and ET to classify materials into different bond types. PCMs fall into the green region of the map, corresponding to metavalent bonding. [14]

2.2 PCM Superlattices

A superlattice (SL) is a periodic structure composed of alternating layers of two or more different materials, each with a precisely controlled thickness on the nanometre scale. This engineered layering modifies electronic, optical, and thermal properties, enabling advanced functionalities in semiconductors, thermoelectrics, and phase-change materials. [15]

In 2011 Simpson et al. demonstrated that superlattice structures of GeTe/Sb₂Te₃ outperform polycrystalline GST in terms of power consumption, endurance, and speed. These superlattices consist of epitaxially matched alternating thin layers of GeTe and Sb₂Te₃ stacked along the [001] direction, forming a periodic bilayer structure. They proposed a novel switching mechanism, which unlike conventional PCMs that rely on melt-quenching and recrystallization, achieves switching through local atomic rearrangement at the GeTe/Sb₂Te₃ interface. This process is electrically driven rather than thermally induced, significantly reducing energy consumption and improving switching speed. [4]

Despite their promising performance [16, 17], superlattice phase-change materials remain at the centre of an ongoing and highly contested debate regarding their switching mechanism. While some studies support an interface-driven, non-thermal process [5, 15, 17], others continue to argue that conventional melt-quenching remains the dominant pathway [18-20]. This unresolved controversy highlights a fundamental gap in our understanding of SL operation. Addressing this gap by investigating the microscopic origins of switching in well-controlled superlattice systems is a central objective of this thesis.

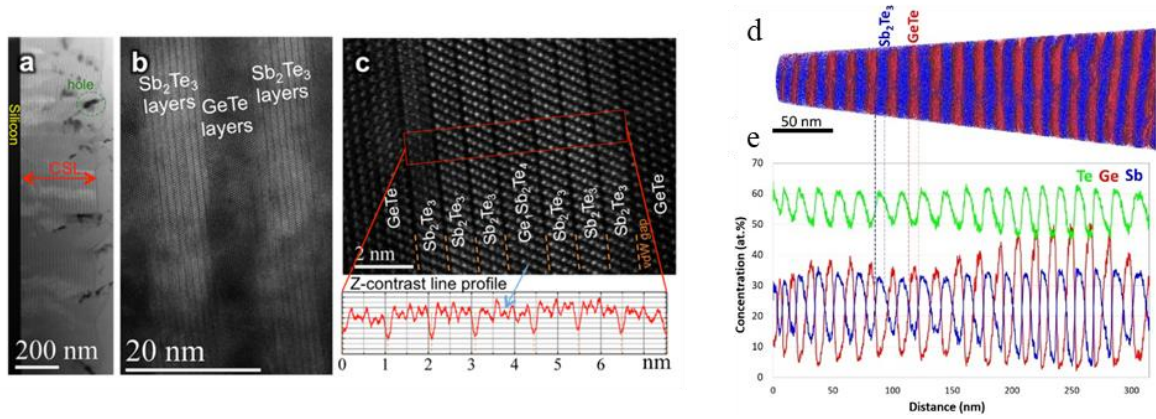


Figure 2.5: HAADF STEM micrographs of CSL layer sputter deposited on Si (1 1 1) substrate. The micrographs were acquired at (a) low magnification, (b) medium magnification, and (c) high-resolution. The Sb₂Te₃ regions are clearly identifiable due to their higher atomic number (Z) contrast and the presence of van der Waals (vdW)-like gaps, as highlighted in (b) and (c). In (c), these gaps are additionally marked with orange dashed lines. The red box in (c) indicates the region used for the Z-contrast line profile analysis. (d) 3D elemental map of Ge (red), Sb (blue) and Te (green) and (e) corresponding 1D profiles along the complete APT tip. [21]

Cojocaru-Mirédin et al. investigated the chemical composition of GeTe/Sb₂Te₃ SLs fabricated by sputter deposition using a correlative transmission electron microscopy-atom probe tomography (TEM-APT) approach. The study found a dominance of Sb towards the

van der Waals (vdW)-like gaps, with cation impurities ranging between 4 to 7 at.% Ge in the Sb_2Te_3 layer and 5 to 13 at.% Sb in the GeTe layer. This intermixing resulted in the formation of GST blocks, with TEM images (Figure 2.5) revealing both $\text{Ge}_1\text{Sb}_2\text{Te}_4$ and $\text{Ge}_2\text{Sb}_2\text{Te}_5$ compounds. [21]

More recently Cojocaru-Mirédin et al. studied the intermixing and ordering phenomena in GeTe/ Sb_2Te_3 crystalline superlattices (CSLs) under heat treatment. Significant intermixing was observed in the pristine state, with heat treatment resulting in stable $\text{Ge}_2\text{Sb}_2\text{Te}_5$ and $\text{Ge}_3\text{Sb}_2\text{Te}_6$ lamellae even after prolonged annealing at 350 °C. This transformation is attributed to the preservation of van der Waals -like gaps within the phase change material layer, which stabilize the $\text{Ge}_3\text{Sb}_2\text{Te}_6$ phase by maintaining structural integrity and preventing excessive interdiffusion. Finite element modeling suggests that this phase evolution during annealing enhances electrothermal properties by reducing RESET current. [22]

The superior performance of SLs was also demonstrated by Kahn et al. as shown in Figure 2.6. The plot clearly illustrates that SLs require significantly lower RESET current for amorphization and show a higher resistance difference between SET and RESET states. They hypothesize a thermally driven phase change mechanism, attributed to electro-thermal confinement effects in the van der Waals-like interfaces of the SL. [7]

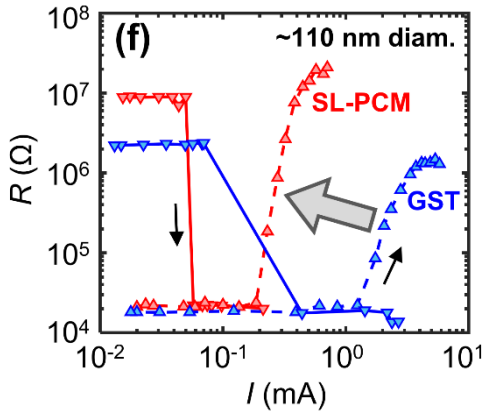


Figure 2.6: Resistance vs. current comparison of SL and GST devices. The RESET current shows up to $\sim 10\times$ reduction in the SL device. [7]

Based on the findings from the aforementioned studies, it can be concluded that optimizing the phase-change properties of SLs requires precise growth techniques. Molecular Beam Epitaxy (MBE) enables the fabrication of high-quality superlattices with controlled stoichiometry and well-defined interfaces. [23] In this work we will investigate high quality, MBE-grown $\text{Ge}_1\text{Sb}_2\text{Te}_4/\text{Sb}_2\text{Te}_3$ superlattices, which were grown by Lucas Bothe during his Master's thesis. [24] $\text{Ge}_1\text{Sb}_2\text{Te}_4$ is chosen as the lowest stable configuration of GST compounds to avoid intermixing.

2.3 Amorphization and Crystallization mechanisms in PCMs

Phase change materials (PCMs) enable reversible transitions between amorphous and crystalline phases, typically induced by laser pulses. The switching mechanism fundamentally relies on the thermal response of the material, where the characteristics of

the laser pulse, such as focus area, power, and duration, control whether the material undergoes crystallization or amorphization. This dependence is succinctly captured in the Time-Temperature-Transformation (TTT) diagram, a central tool for understanding the kinetics of PCM switching. [25]

Time Temperature Transformation (TTT) Diagram

In its typical form (Figure 2.7), the TTT diagram delineates the stable and metastable regions of PCM behavior and defines boundaries between amorphous, supercooled liquid, and crystalline phases.

In phase change materials, the thermal behavior is governed by three key temperatures: the glass transition temperature (T_g), the crystallization temperature (T_C), and the liquidus temperature (T_L). The liquidus temperature T_L defines the transition from a fully liquid state to a solid-liquid equilibrium in alloys like PCMs, even though the melting temperature T_m , which is strictly defined for pure elements, is more commonly used in PCM literature to describe the onset of amorphization via melt-quenching. It also has to be noted that although T_g marks the transition to an amorphous state, T_C is more commonly used in PCM studies. Above T_L , the material is fully molten, while the region between T_L and T_C defines the supercooled liquid state, disordered but still capable of rapid crystallization. This temperature range is critical, as small variations in time and temperature determine whether the material crystallizes or remains amorphous. [25] Depending on the trajectory taken through this diagram, four fundamental transformation processes are typically identified:

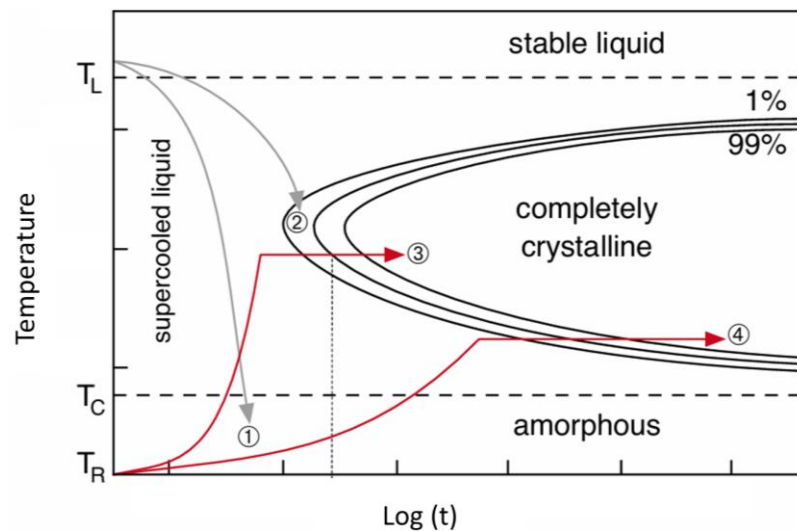


Figure 2.7: Schematic of a time-temperature transformation diagram for a PCM with four numbered phase changes. The liquidus (T_L), crystallization (T_C) and room (T_R) temperatures are marked. The diagram shows representative phase transition processes, including melt-quenching (1, 2) and thermal annealing (3, 4), distinguished by their respective cooling and heating rates. The amorphous, crystalline, and liquid states of the PCM are indicated. Solid curves correspond to crystallization degrees of 1%, 50% (central curve), and 99%. [26]

Ultrafast Quenching - Amorphization

In this regime, the PCM is heated above T_L into the liquid phase and then cooled extremely rapidly, with quenching rates typically exceeding 10^9 – 10^{11} K/s. [27] The cooling path passes vertically downwards in the TTT diagram, completely bypassing the crystallization "nose." Because the atomic mobility becomes vanishingly small before atoms can organize into a crystalline structure, the material is "frozen" in a disordered, liquid-like configuration, forming a glassy (amorphous) state. This state is metastable at room temperature, meaning it is not in thermal equilibrium but can persist for over a decade at 80 °C, an essential property for long-term data retention in PCM memory devices. [28] The key characteristic of this region is the complete suppression of nucleation and crystal growth, driven by the lack of time for rearrangement and the abrupt loss of mobility below T_c .

Insufficient Quenching - Partial Crystallization

This path also starts with heating the PCM into the liquid phase ($T > T_L$), but the cooling rate is slower, falling short of the critical quenching rate. The trajectory grazes or intersects the crystallization contour in the TTT diagram. In this case, atomic mobility remains sufficient long enough for nucleation and partial growth of crystalline regions before the structure becomes frozen. The result is a mixed-phase material, largely amorphous but containing embedded crystalline grains. Such partial crystallization negatively impacts the optical and electrical contrast required for reliable switching and is thus an undesirable intermediate state in memory applications. This process highlights the sensitivity of PCM phase outcomes to thermal management, emphasizing the need for precise control of pulse duration and cooling rates.

Low-Temperature Annealing - Slow Crystallization

Starting from the amorphous phase ($T < T_c$), this process involves moderate heating into the temperature range between T_c and T_L , with the material annealed at a relatively low temperature for an extended time. Although the atomic mobility at these temperatures is limited, it is non-zero, and over time, atoms can slowly rearrange to nucleate and grow crystalline domains. The slope of this process line in the TTT diagram is shallow, indicating slow kinetics. This type of transformation is relevant in long-term drift or aging phenomena: even at modest operating temperatures, amorphous PCMs can spontaneously crystallize over months or years, potentially degrading data integrity. Such drift must be carefully considered in memory device design and reliability testing.

High-Temperature Annealing - Fast Crystallization

This annealing process also begins in the amorphous phase but occurs at higher temperatures within the supercooled liquid regime, closer to T_L . The increased temperature

accelerates atomic mobility, enabling rapid nucleation and crystallization. In the TTT diagram, this is visualized as a faster slope, indicating shorter transformation times. This mechanism is exploited in on-switching operations in PCM-based memory, where a brief heating pulse induces crystallization within nanoseconds. Experimental observations have confirmed that crystallization from the supercooled liquid can occur on such short timescales, though direct measurement remains challenging. Crystallization in this regime can yield different crystalline phases for a PCM, depending on the thermal pathway, potentially influencing electrical and optical properties.

In summary, the TTT diagram encapsulates the delicate balance between temperature, time, and atomic mobility that governs phase transitions in PCMs. Understanding this diagram is crucial for data storage applications, ensuring reliable and reversible switching between amorphous and crystalline states.

Nucleation and growth-dominated crystallization

Crystallization is a complex and fundamentally slower process; it involves two concurring and competing processes: nucleation and growth. [25] Nucleation refers to the spontaneous formation of small crystalline clusters within the amorphous matrix. (Figure 2.8 b) In the case that the cluster is stable, it can grow to a macroscopic size. Growth then refers to the progression of the phase front separating the amorphous and crystalline regions into the amorphous PCM volume. (Figure 2.8 b) The thermodynamics of this transformation hinge on a balance between volume energy, which favors crystallization, and interfacial energy, which resists it. This competition defines a critical nucleus radius r_c , clusters smaller than r_c dissolve, while those larger can grow. As such, crystallization is inherently probabilistic, and nucleation is often preceded by an incubation period, ranging from microseconds in as-deposited films to nanoseconds in partially crystalline regions. [29]

Temperature plays a central role in both nucleation and growth rates, but importantly, their maxima do not coincide. At lower temperatures, nucleation dominates, as atomic mobility is sufficient to form nuclei but too limited for substantial growth. At higher temperatures, closer to the melting point, mobility increases and favors rapid growth, though nucleation may decline. (Figure 2.8 a) This gives rise to two classes of PCM behavior: nucleation-dominated systems like $\text{Ge}_2\text{Sb}_2\text{Te}_5$ (GST), where many small nuclei form and grow slowly, and growth-dominated systems like AgInSbTe , where few nuclei form but grow rapidly. [28] The dominant mechanism significantly impacts switching speed and device performance. For instance, growth-dominated materials can recrystallize in nanoseconds, making them advantageous in memory devices.

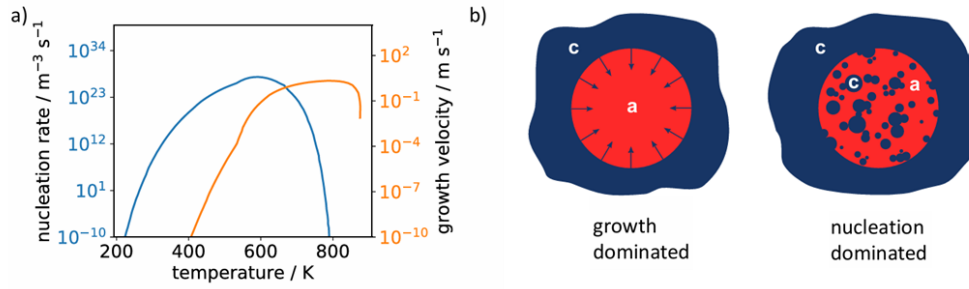


Figure 2.8: a) Plot of the nucleation rate and growth velocity of crystalline grains of GST-225 against the temperature [30], b) Crystallization mechanisms of an amorphous region within a crystalline matrix: on the left, growth-dominated crystallization begins at the phase boundary and proceeds inward, on the right, nucleation-dominated crystallization features randomly distributed nuclei forming within the amorphous area and growing outward toward the boundary [31].

The difficulty of crystallization therefore lies primarily in its dependence on sustained thermal conditions and precise atomic rearrangement. A sufficient temperature must be maintained for a critical duration to allow for the formation and propagation of crystalline order. Unlike amorphization, which is primarily a thermal quench event, crystallization demands a carefully balanced heating regime that encourages order without exceeding the melting point. As temperature increases toward it, growth rate increases, but nucleation rate drops. Above the melting point, growth becomes negative due to reamorphization. This leads to a delicate trade-off: the optimal switching window must ensure enough nucleation and controlled growth without exceeding the melting point. The crystallization window is inherently narrow and asymmetric, since nucleation and growth are not maximized at the same temperature. [30]

Laser-induced switching of PCMs

In optical switching of PCMs, both amorphization and recrystallization are driven by carefully tailored laser pulses that exploit the material's thermally activated phase transition behavior. (Figure 2.9) Crystallization is induced when an amorphous PCM is locally heated above the glass transition temperature T_g , but below the melting temperature T_m , allowing atoms sufficient mobility and time to rearrange into a thermodynamically stable crystalline structure. This is typically achieved with laser pulses of relatively low energy and longer duration, ensuring controlled heating within the crystallization window.

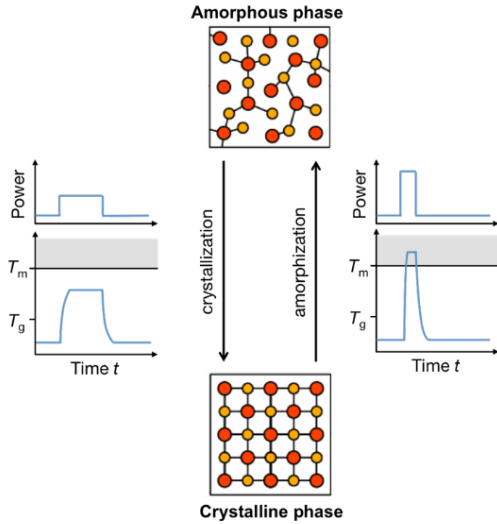


Figure 2.9: Reversible switching in phase change materials: Crystallization occurs through a long, low-power pulse that heats the material above the glass transition temperature (T_g) but remains below the melting temperature T_m . In contrast, amorphization is triggered by a short, high-intensity pulse that raises the temperature above T_m , followed by rapid cooling.[31]

On the other hand, amorphization requires a fundamentally different thermal approach. A crystalline PCM must first be driven above its melting temperature T_m , transitioning it into a highly mobile liquid phase. To lock in the disordered atomic configuration characteristic of the amorphous state, the molten material must then be rapidly quenched to suppress crystallization. This is accomplished by a high-energy, ultra-short laser pulse that ensures both complete melting and rapid cooling, freezing the material into a metastable glassy state. Together, these two pulse regimes, long and moderate for crystallization, short and intense for amorphization, form the operational basis for ultrafast and localized optical switching in PCMs.

Ultimately, while both switching directions rely on thermal manipulation, crystallization is more demanding due to its reliance on atomically precise self-organization over time, compared to the relatively straightforward nature of quenching a melt into disorder. This asymmetry in switching kinetics represents a fundamental bottleneck in PCM-based memory technologies, particularly when targeting ultra-fast write/erase cycles.

FABRICATION OF $\text{Ge}_1\text{Sb}_2\text{Te}_4$ / Sb_2Te_3 SUPERLATTICES

3.1 Molecular Beam Epitaxy (MBE)

During the course of this work a series of superlattices made of two individual materials $\text{Ge}_1\text{Sb}_2\text{Te}_4$ and Sb_2Te_3 , will be used. The superlattices were grown in an MBE chamber of the Nanocluster at the Helmholtz Nano Facility (HNF) at Forschungszentrum Jülich (FZJ). Highly textured thin films can be grown in an Ultra High Vacuum (UHV) environment in MBE by evaporating pure elements in effusion cells. The evaporated atoms from the effusion cells condense onto the heated substrate, resulting in the formation of epitaxial thin films. The deposition via effusion cells makes it possible to control the fluxes to extremely low values. By using low deposition rates, the film thickness can be controlled precisely allowing for the creation of abrupt material interfaces and making it particularly well-suited for the fabrication of heterostructures such as superlattices. The growth process can be monitored using a Reflection High-Energy Electron Diffraction (RHEED) setup, which includes a RHEED gun and a RHEED screen. [32] The main parts of an MBE chamber are shown in Figure 3.1.

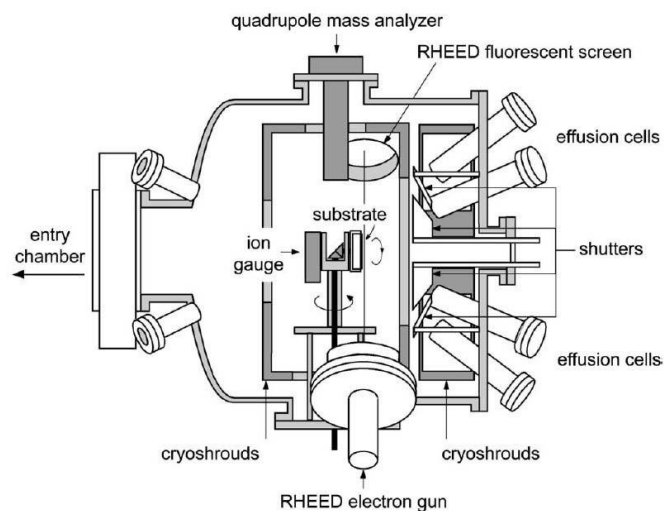


Figure 3.1: Schematic of an MBE chamber. [33]

3.2 Sb_2Te_3 and $\text{Ge}_1\text{Sb}_2\text{Te}_4$

Antimony telluride (Sb_2Te_3) is a phase change material, a narrow-gap semiconductor, and a topological insulator. [34, 35] It crystallizes in a rhombohedral structure (space group $\bar{R}3m$) with a layered arrangement along the [0001] direction. The material consists of quintuple layers (QLs) of -Te-Sb-Te-Sb-Te-, which are stacked with van der Waals-like (vdW-like) gaps between them. Three QLs make up one hexagonal unit cell.

$\text{Ge}_1\text{Sb}_2\text{Te}_4$ (GST-124) has a hexagonal atomic structure with a unit cell composed of three septuple layers arranged in an ABC-stacking sequence. These septuples are separated by vdW-like gaps between the outermost Te atoms, similar to Sb_2Te_3 . This similarity is expected, as GST-124 represents the lowest configuration in the pseudobinary line between GeTe and Sb_2Te_3 . Upon crystallization from the amorphous phase, GST-124 initially forms a metastable rocksalt structure before transitioning to its stable hexagonal phase. (Figure 3.2)

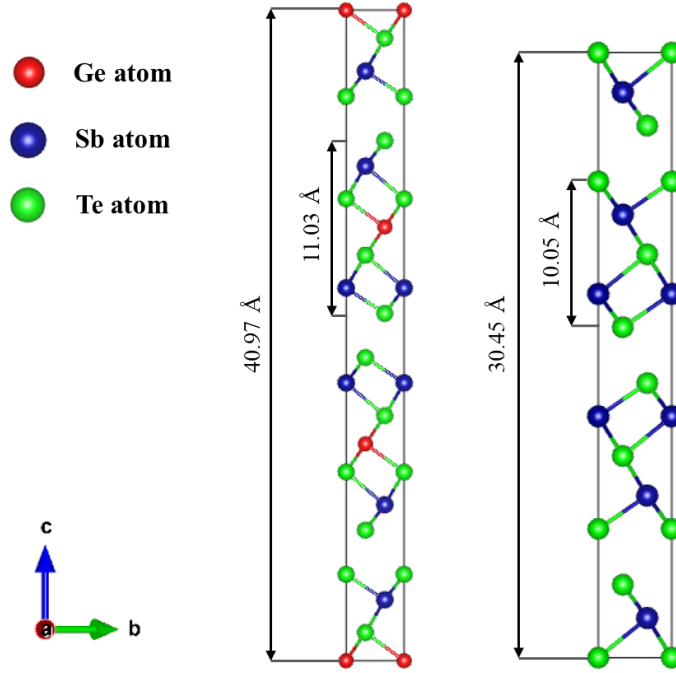


Figure 3.2: Structure of Sb_2Te_3 (left) and GST-124 (right), made with VESTA.

3.3 Fabrication of the samples

The following part is mainly explained using results and analysis from [24], which also includes the fabrication and structural characterization of the samples used in this work. The Sb_2Te_3 /GST-124 superlattices examined in this work were fabricated using Molecular Beam Epitaxy (MBE), applying previously optimized growth conditions for each material.

The deposition process began on a Te-passivated Si(111)-1×1 surface to allow vdW-epitaxy, at a substrate temperature of 145 °C, with a targeted total thickness of 60–65 nm for each superlattice. The first layer was Sb₂Te₃, followed by alternating layers of GST-124 and Sb₂Te₃. Due to its better oxidation resistance, Sb₂Te₃ was also chosen as the final layer of each structure. A one-minute growth interruption was introduced between layers to improve stoichiometric control and allow incomplete layers to finish forming, thereby limiting intermixing between the constituents. Once the growth was completed, the samples were cooled down in the MBE chamber and cut into smaller sections for further processing. For specific experiments, such as optical switching, some samples were capped with approximately 25 nm of sputtered ZnS:SiO₂. This capping enhanced laser absorption during optical tests. ZnS:SiO₂ could later be removed using HF-dipping without damaging the superlattice. In addition, some samples were capped with 200 nm of indium as a protective layer for the fabrication of needle-shaped specimens used in atom probe tomography (APT) measurements.

The N–X series of superlattices was designed with two GST-124 septuples in each supercell, and an increasing number of Sb₂Te₃ quintuples from 2 to 8. (APPENDIX, Figure 7.1) This configuration enabled a controlled study of how the layered architecture affects superlattice behaviour, particularly regarding optical switching.

Superlattice (N:X)	GeSb ₂ Te ₄ quintuples per supercell	Sb ₂ Te ₃ septuples per supercell	Repetitions of supercell	Total thickness (nm)
N:2	2	2	12	61
N:4	2	4	8	67
N:6	2	6	6	72
N:8	2	8	4	63

Table 1: Information about the N:X Superlattice series

The structural quality of the as-deposited Sb₂Te₃/GST-124 superlattices was evaluated using $\Theta/2\Theta$ X-ray diffraction scans. Initially, key features were identified directly from the measured diffraction patterns. This was followed by detailed fitting of the scans to extract parameters such as superlattice stoichiometry and the periodicity of the superlattice unit cell, Λ_{sl} .

The $\theta/2\theta$ scans displayed in Figure 3.3 are characterized by distinct peaks originating from the superlattice supercell, formed by a single GST-124 septuple and an Sb₂Te₃ quintuple. Due to the large periodicity of this supercell relative to the individual materials' unit cells,

the diffraction pattern features a dense series of superlattice peaks. However, structural dephasing and varying structure factors prevent many of these from being visible experimentally. Among the observable peaks, an artificial classification into "main peaks" and "satellite peaks" is applied, based on intensity differences that arise from structural factors, despite all peaks belonging to the same diffraction family. Satellite peaks are identified as lower intensity features adjacent to the dominant main peaks.

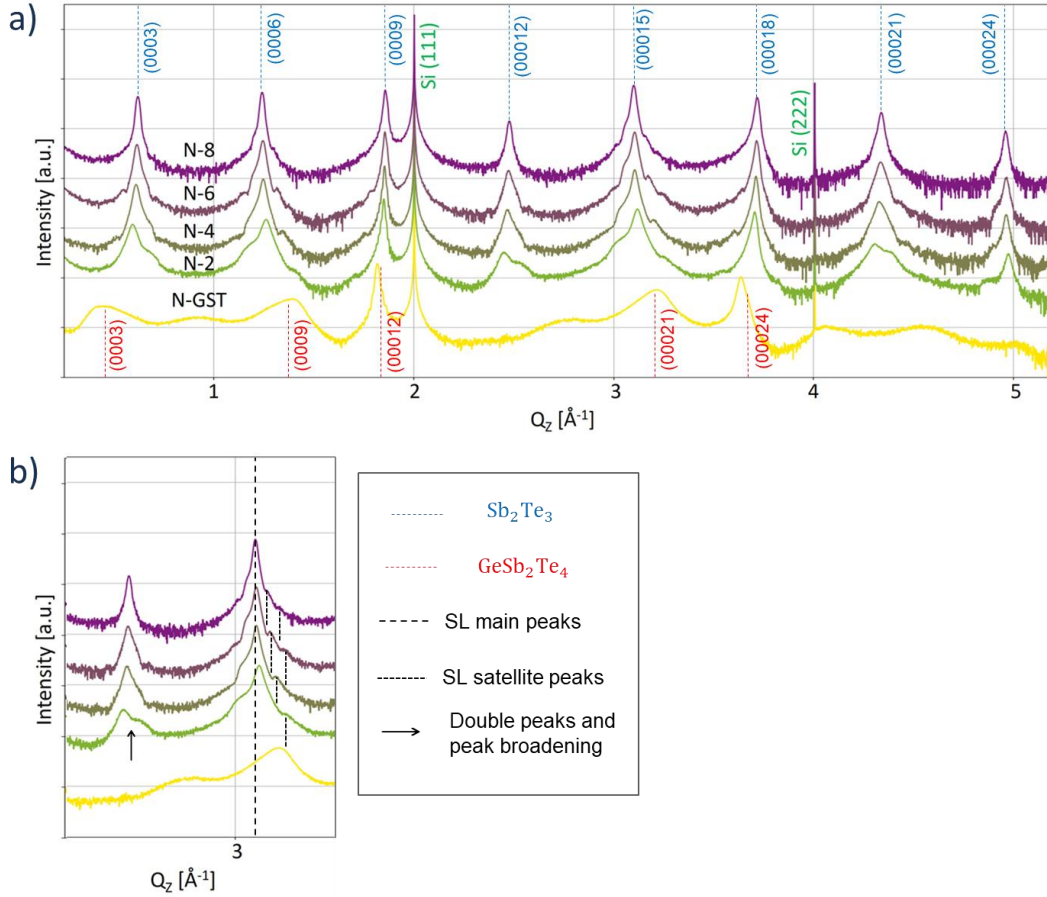


Figure 3.3: a) $\theta/2\theta$ -scans of the N-X superlattices with the reference GST-124 sample. Vertical dashed lines show (000l)-peak families for Sb_2Te_3 and GeSb_2Te_4 . b) Region around the second superlattice peaks group around $Q_z = 3.1 \text{ \AA}^{-1}$. SL peak groups are found around positions where peaks of GST-124 and Sb_2Te_3 are close to each other.

Across the full N-series, typical superlattice signatures, such as recurring main and satellite peaks, are consistently observed. Additionally, structural features associated with Sb_2Te_3 , including peak broadening and double peaks near $Q_z = 2.47 \text{ \AA}^{-1}$ and $Q_z = 4.33 \text{ \AA}^{-1}$, appear more prominently with increasing Sb_2Te_3 content. As expected, the SL-peaks gradually shift toward the characteristic peak positions of bulk Sb_2Te_3 as its proportion increases, with the N-8 sample showing near coincidence between SL and Sb_2Te_3 peaks.

Overall, the structural investigation confirms that high-quality superlattices were successfully fabricated in the N-series, with good alignment between nominal growth parameters and experimentally determined structural properties for the overall and supercell thickness, (Figure 3.4).

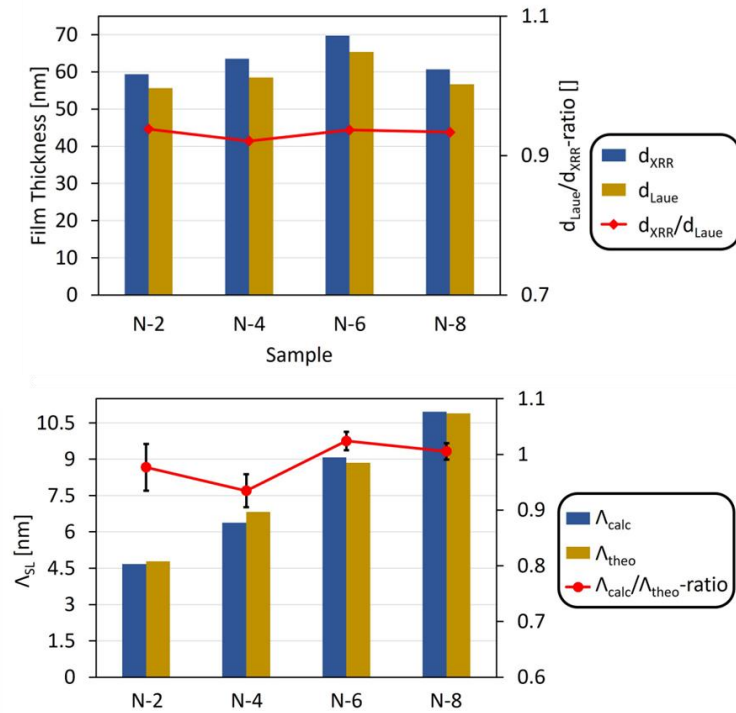


Figure 3.4: (Top) Film thickness computed from XRR measurements (d_{XRR}) and crystal thickness calculated from the Laue oscillations (d_{Laue}). Included is the $d_{\text{XRR}}/d_{\text{Laue}}$ ratio. (Bottom) Desired and calculated superlattice supercell thickness Λ_{SL} . Included is the $\Lambda_{\text{calc}}/\Lambda_{\text{theo}}$ ratio.

EXPERIMENTAL BASICS: CHARACTERIZATION TECHNIQUES OF PCM SUPERLATTICES

4.1 Atom Probe Tomography (APT)

Atom probe tomography (APT) is an advanced microscopy technique that allows for the investigation of chemical composition with near-atomic precision. It offers three-dimensional imaging capabilities that reveal the microstructure of materials in exceptional detail. One of the key innovations of APT is its ability to differentiate materials based on their bonding mechanisms, providing unique insights into their atomic-scale properties. [36]

Atom Probe Tomography (APT) evolved from key advances in field emission and ion microscopy, beginning in the 1950s when Erwin Müller used high electrostatic fields to sharp metal tips and was able to image individual atoms via Field Ion Microscopy (FIM). [37] This technique was later combined with time-of-flight mass spectrometry to create the first atom probe, enabling both spatial imaging and chemical identification of atoms as they were field-evaporated from a specimen. [38]

In the decades that followed, APT advanced rapidly. Innovations like laser pulsing enabled analysis of non-conductive materials, significantly broadening APT's applicability. The introduction of position-sensitive detectors culminated in the 3-D atom probe (3-DAP), capable of mapping atomic positions and identities with sub-nanometre precision. The greatest advancement came with the development of the Local Electrode Atom Probe (LEAP), which dramatically increased data collection rates (to millions of atoms per minute), mass resolving power, and field of view. [39]

Today's LEAP systems, represent the culmination of decades of technological progress, having the highest spatial resolution among analytical techniques (0.2nm), and the highest absolute analytical sensitivity (single atoms), APT is one of the most powerful tools for 3-D compositional analysis at the atomic scale.

The following sections are taken from Larson et al. [40], Gault et al. [41] and Kelly et al. [42].

4.1.1 Working Principle

Atom probe tomography (APT) relies on field evaporation, a process in which a high electric field at the surface of a sharply pointed specimen ionizes atoms, even at cryogenic temperatures. Field strengths on the order of 10^{10} V/m strip electrons from surface atoms via tunneling, creating positively charged ions that are accelerated toward a detector. This field is generated by applying a high voltage of several kV to a specimen with a tip radius typically below 100 nm, where the electric field strength is inversely proportional to tip radius, making specimen sharpness essential. Ions are evaporated in a controlled manner, usually at a rate of one ion per hundred pulses, to minimize multiple hits and reduce stress on the sample.

An APT system consists of a specimen mount, a counter electrode with an aperture, and a position-sensitive detector. (Figure 4.1) The analysis is conducted in an ultra-high vacuum (10^{-8} to 10^{-9} Pa), with the specimen cooled to cryogenic temperatures (20–80 K). Field evaporation is initiated by either voltage pulses (for conductive materials) or laser pulses (for a broader range of materials, including semiconductors and insulators). Laser pulses, typically in the pico- to femtosecond range, are focused on the specimen apex to induce brief thermal excitation.

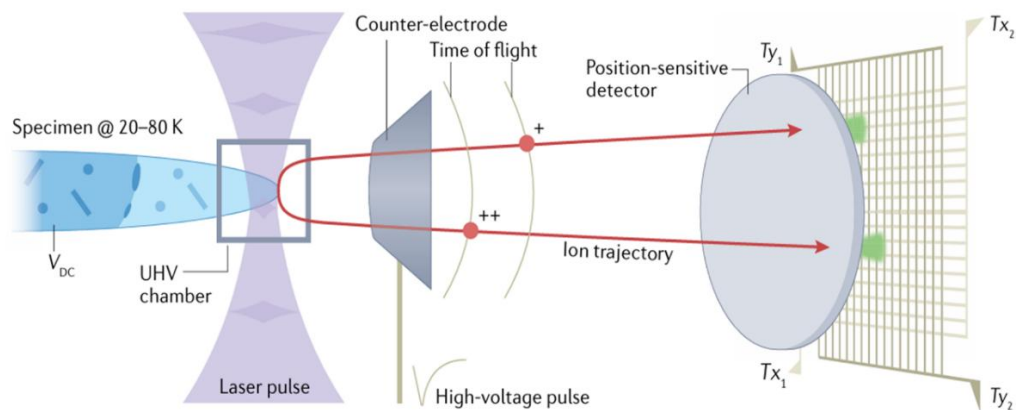


Figure 4.1: Schematics of the atom probe instrumentation. [43]

Since evaporation occurs atom by atom, the sequence of detected ions enables reconstruction of the third spatial dimension. The projection from the ~ 100 nm tip onto a ~ 100 mm detector yields a magnification of about one million, resolving interatomic distances at the atomic scale. A delay-line detector records both the impact position and

time-of-flight (TOF) of each ion with high precision, allowing determination of each atom's original position and identity. The resulting 3D dataset offers a detailed atomic map of the material's composition and structure.

4.1.2 Field Evaporation

Field evaporation is the key physical process that makes atom-by-atom analysis possible in APT. In this mechanism, atoms are stripped from the surface of a material as positively charged ions when exposed to a high electric field, typically in the range of 10^9 to 10^{10} V/m, remaining an electron inside the specimen. (Figure 4.2)

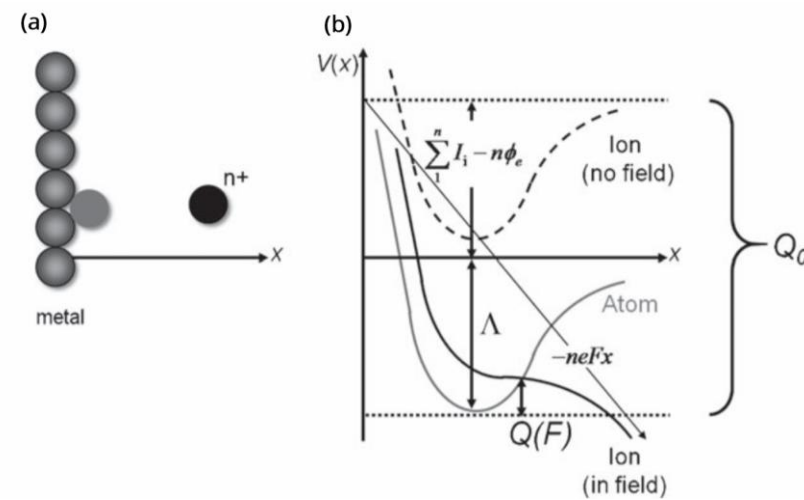


Figure 4.2: (a) Schematic of the field evaporation process: an atom is ionized and subsequently emitted from the specimen surface, while its corresponding electron remains within the material. (b) Potential energies of an atom and an ion with and without an external electric field. The applied field reduces the energy barrier $Q(F)$ that must be overcome for ionization and evaporation to occur. [41]

To create such extreme field strengths, a high voltage is applied to a needle-shaped specimen with an apex radius smaller than 100 nm. The tip is commonly modelled as a truncated cone capped with a hemisphere. This shape is defined by two parameters: the shank angle, which characterizes the cone's taper, and the apex radius of the curvature. The electric field at the tip can be approximated by:

$$F = \frac{V}{k_f \cdot R} \quad (4.1)$$

In this equation, V is the applied voltage, R is the apex radius, and k_f is a shape-dependent field factor that captures the influence of geometry and surrounding electrostatic conditions. For metallic specimens like tungsten, k_f typically falls between 3 and 8. Sharper tips or nearby conductive features, such as a flat mounting base, tend to reduce the effective field factor, thereby increasing the local electric field.

To stabilize surface atoms and prevent their unwanted motion, the specimen is cooled to cryogenic temperatures, usually between 20 K and 80 K. This minimizes atomic vibrations and significantly slows surface diffusion, helping to maintain the spatial integrity of atomic positions during the evaporation process.

To regulate the evaporation of atoms in a controlled and time-resolved manner, two primary approaches are employed: either the electric field is intensified at a fixed temperature, or the temperature is elevated while maintaining a constant field. These methods correspond to the two fundamental operational modes of APT, high voltage pulsing and laser pulsing, respectively.

In the high-voltage (HV) pulsing mode, a series of short voltage pulses is superimposed on a steady bias, temporarily raising the local electric field above the threshold required for evaporation. This approach takes advantage of the reduced activation barrier that comes with increased field strength and is especially effective for conductive materials.

The alternative, and more versatile, laser-pulsing mode, relies on thermal excitation. Here, the specimen is held under a high DC bias, and brief laser pulses, typically in the range of a few picoseconds, are directed at the tip apex. These pulses induce a rapid, localized rise in temperature, providing the necessary energy to overcome the evaporation barrier without changing the applied field. In this work, the laser-pulsing mode is employed.

Laser pulsing offers several benefits in APT. First, because ion emission occurs only after the arrival of a laser pulse, and acceleration is driven solely by the static field, mass resolution is significantly improved compared to HV pulsing. Second, this method broadens the range of materials that can be analysed, enabling the study of semiconductors, oxides, dielectrics and other non-conducting phases. However, a trade-off exists: the increased thermal energy can enhance surface diffusion, slightly reducing spatial resolution due to lateral atomic movement prior to evaporation.

4.1.3 Time of Flight Mass Spectrometry

Time-of-Flight (ToF) mass spectrometry plays a crucial role in Atom Probe Tomography (APT) by enabling the identification of individual atoms through measurement of their flight times. This technique provides a direct link between the time an ion takes to reach the detector and its mass-to-charge state ratio (m/n), allowing for precise chemical analysis at the atomic scale.

In APT, atoms are ionized and field-evaporated from the apex of a needle-shaped specimen. This process is initiated by a pulse, which defines the start time (time zero) for the time-of-flight measurement. This is achieved by pulsing either the electric field (HV pulsing) or the specimen temperature (laser pulsing), as mentioned before. Once emitted, the ions are accelerated by a strong electric field and travel toward a position-sensitive detector.

Because the field near the specimen apex is extremely high, ions gain most of their velocity almost immediately after emission. After this rapid acceleration phase, they coast at a nearly constant velocity along the flight path. Assuming the ion reaches this terminal velocity almost instantly, a basic energetic model equates the ion's kinetic energy to its gained electric potential energy:

$$\frac{1}{2}mv^2 = neV \quad (4.2)$$

Substituting $v = \frac{L}{t}$ (with L being the distance to the detector and t the flight time), we arrive at:

$$\frac{m}{n} = \frac{2eV}{L^2}t^2 \quad (4.3)$$

This relationship provides a straightforward way to determine the mass-to-charge ratio of ions based on their time-of-flight, assuming known values for voltage and flight path length.

The effectiveness of ToF mass spectrometry is often judged by its mass resolving power (MRP), which refers to how well the instrument can distinguish between two species with similar mass-to-charge ratios. A higher MRP means better separation of closely spaced peaks in the mass spectrum, leading to more accurate identification of elements and isotopes.

MRP is defined as:

$$MRP = \frac{m}{\Delta m} = \frac{t}{2\Delta t} \quad (4.4)$$

and using equation (4.3) we obtain

$$MRP = \frac{m}{\Delta m} = \frac{t}{2\Delta t} = \frac{t}{2\Delta t} \sqrt{\frac{m}{n} \frac{1}{2eV}} L \quad (4.5)$$

This more complete expression for MRP that accounts for experimental variables expression shows that MRP improves with longer flight times and more precise timing.

Specifically, it highlights how crucial timing precision, flight path length, and operating voltage are in determining the system's resolving capabilities.

In practical terms, this means that peaks corresponding to higher m/n values are generally better resolved, while increasing the accelerating voltage (V) tends to degrade resolution. Additionally, MRP scales with the flight path length L , making system geometry an important design consideration.

The resolution of ToF mass spectrometry is limited by several types of timing uncertainty:

Physical timing uncertainty (δt_p) – arises from variations in voltage and flight path length.

Electronic timing uncertainty (δt_T) – due to the finite resolution of the detector and timing electronics.

Time-of-departure spread (δt_D) – introduced by the finite temporal width of the ionization event during pulsing.

These uncertainties can be combined as:

$$\delta t = \delta t_p^2 + \delta t_T^2 + \delta t_D^2 \quad (4.6)$$

Even under ideal conditions, peak broadening in mass spectra can occur. Particularly during voltage pulsing, not all ions are emitted at the exact same point during the pulse's rise and fall. This leads to asymmetric peak shapes, typically skewed toward higher m/n values due to variations in energy gain.

4.1.4 Probability of multiple events and probability of molecular ions

While APT is a technique commonly used to map elemental distributions, its ability to probe chemical bonding has garnered increasing interest. [36] A key aspect of bond rupture in APT is defined by two probabilities: the Probability of Molecular Ions (PMI) and the Probability of Multiple Events (PME). (Figure 4.3) These metrics provide insight into how chemical bonds respond to laser-assisted field evaporation, allowing for distinctions between different bonding mechanisms. This capability is especially crucial for phase change materials where metavalent bonding plays a central role. [14]

The following sections are taken from Cojocaru et. al [36].

Probability of Molecular Ions (PMI)

The PMI describes the fraction of detected ions that are molecular in nature, rather than monatomic. This parameter varies markedly among different types of materials and is

directly linked to the nature of chemical bonding. Metallic systems, characterized by delocalized electrons and weak directional bonding, typically exhibit negligible PMI values, indicating a strong preference for evaporation as isolated atoms. Materials with stronger and more directional bonding, such as covalent or ionic compounds, demonstrate significantly elevated PMI values. These include oxides, nitrides, and certain chalcogenides, where bonded atom clusters can persist through evaporation and arrive at the detector as intact molecular ions.

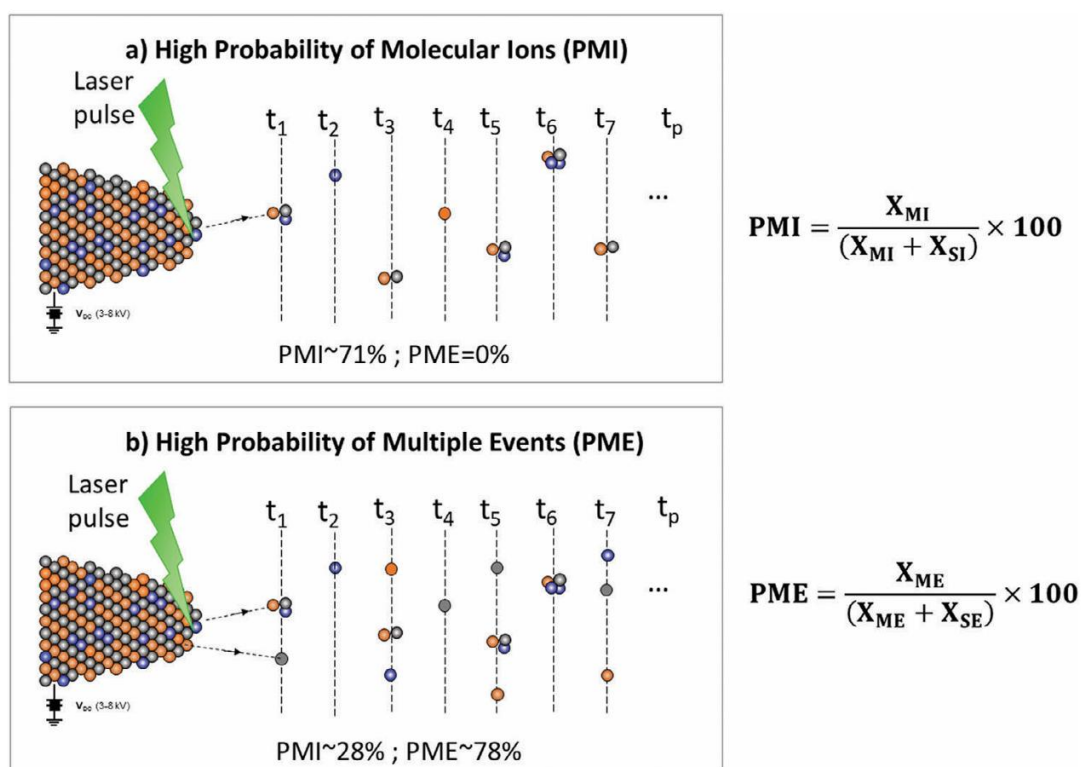


Figure 4.3: Schematic representation of bond-breaking behaviour in atom probe tomography. a) high PMI, indicating the detection of intact molecular ions, and b) high PME, reflecting the simultaneous emission of multiple ions from a single pulse. [36]

In particular, high PMI values can appear for materials exhibiting metavalent bonding (MVB), a non-classical bonding regime that transitions from localized to delocalized electron systems and suggest a distinct field evaporation behaviour arising from partially delocalized electrons and a high bond polarizability. These elevated PMI levels reflect the survival of short-range order and bonded units during field evaporation, enabling the use of PMI as a proxy for local bonding character.

Probability of Multiple Events (PME)

PME quantifies the frequency with which multiple ions are detected in response to a single laser pulse. Empirical data reveal PME values ranging from a few percent in simple metals

to over 90% in complex crystalline materials. Such elevated PME values imply correlated evaporation phenomena and specific bond rupture mechanisms.

Multiple-ion detection can arise from several distinct processes:

1. Correlated field evaporation, in which two or more atoms are ejected simultaneously due to their spatial and chemical proximity.
2. Molecular ion dissociation, whereby a molecular ion emitted from the specimen undergoes in-flight fragmentation into two or more charged species.
3. Surface migration and clustering, which may lead to the emission of atom clusters due to local rearrangement at the specimen apex.

Importantly, PME can serve as a sensitive indicator of non-classical evaporation behavior. For instance, in crystalline MVB compounds, PME values consistently exceed 50%, significantly higher than those observed in covalently bonded semiconductors or metals, which typically lie below 20%.

Combined Interpretation and Classification

Together, PMI and PME offer a multidimensional approach to classify bonding behavior in materials analyzed via APT. Metals, with their high conductivity and short screening lengths, consistently show low PMI and PME values due to rapid and independent ionization events. Covalent materials exhibit low PME but may show moderate to high PMI due to the robustness of bonded units under field evaporation. MVB materials display both high PMI and PME, indicative of their unique bonding mechanism, distinct from purely covalent or metallic bonding, and extended screening lengths that affect the evaporation dynamics.

These trends have been further correlated with intrinsic electronic properties such as electrical conductivity and dielectric response. At the transition between localized (insulating) and delocalized (metallic) electronic regimes, field penetration depths become anomalously large, altering the potential landscape at the specimen surface and resulting in enhanced correlated evaporation and molecular ion formation. Theoretical models of screening length, derived from either Fermi liquid or semiconductor band-structure considerations, support these observations, with the transition point corresponding to a pronounced increase in both PME and PMI.

The quantitative evaluation of PME and PMI thus extends the analytical capability of APT from purely compositional profiling to a deeper understanding of material bonding and electronic structure. These metrics provide critical insights in the study of phase-change materials, semiconductors, thermoelectrics, and complex oxides, where local bonding configurations are often strongly tied to functional properties. Furthermore, by mapping

PME and PMI across three-dimensional reconstructions, it becomes possible to spatially resolve variations in bonding character, enabling the identification of heterogeneous bonding environments within single specimens.

In conclusion, PMI and PME serve not merely as statistical descriptors of evaporation behavior, but as robust physical indicators of the underlying bonding landscape in materials. Their combined analysis allows for the classification of bonding types, the inference of electronic structure features, and the design of targeted APT studies to probe material behavior at the atomic scale.

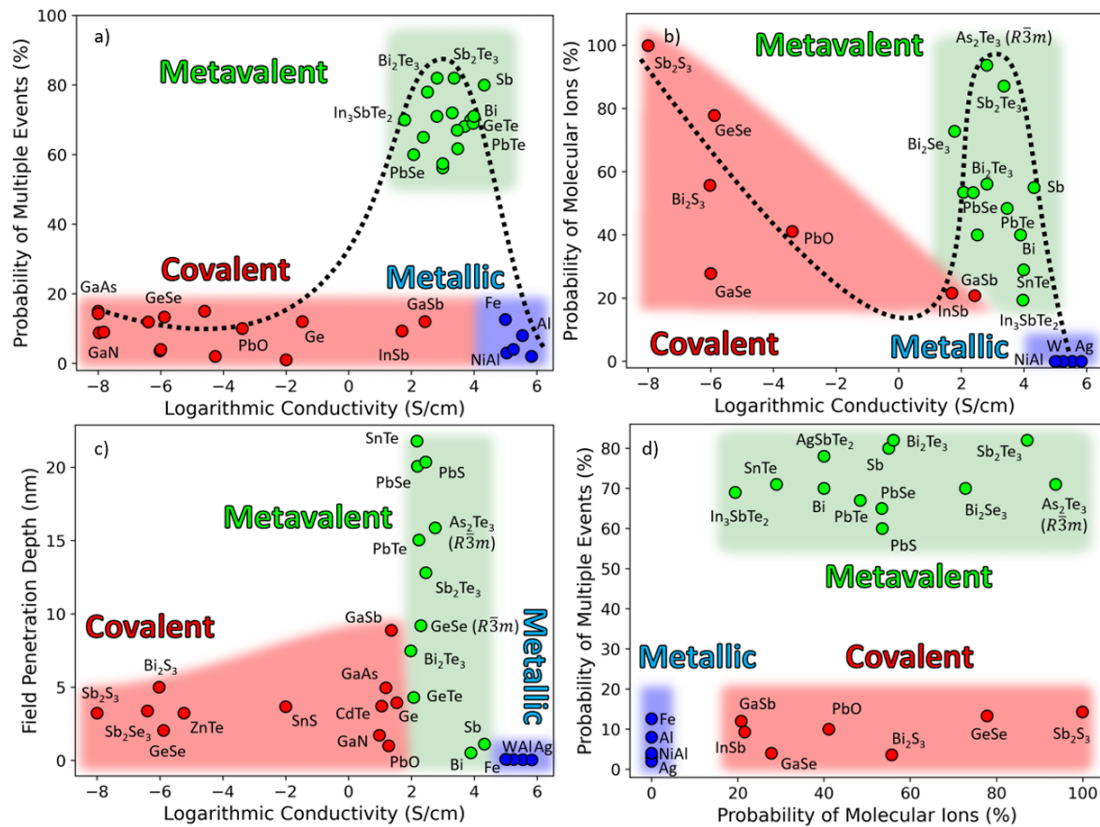


Figure 4.4: a) PMI plotted against conductivity, b) PME plotted against conductivity, c) Field penetration depth plotted against conductivity, d) PMI plotted against PME. Several classes of materials can be distinguished based on their bond breaking in the atom probe: metals (in blue), covalently bonded compounds (in red) and the green region, which marks the transition from localized to delocalized electron systems, where metavalently bonded materials are prominent. [36]

4.1.5 Sample Preparation

As highlighted throughout the previous sections, the sharp, needle-like shape of the sample is central to atom probe functionality, as it influences the strength of the local electric field, the trajectories of the emitted ions, and the resulting mass resolution, making precise sample preparation a fundamental step in any APT analysis. The process utilizes a dual-beam focused ion beam/scanning electron microscope (FIB/SEM) system to extract a wedge-shaped lamella from the material (liftout), which is then milled annularly into a sharp, conical tip suitable for atom probe analysis, as depicted in Figure 4.5.

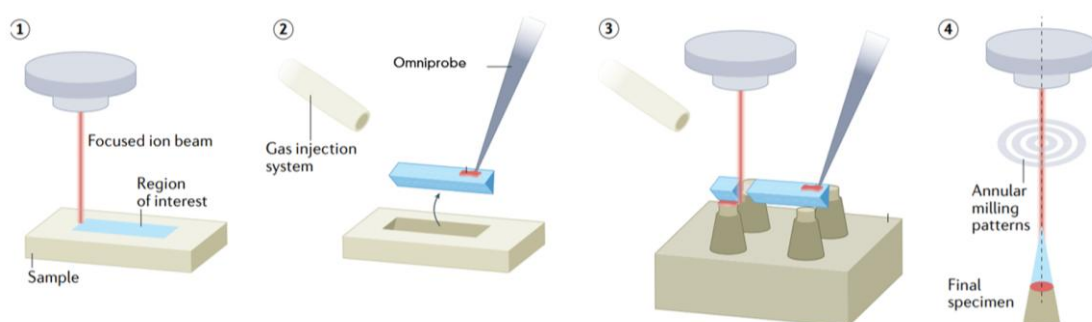


Figure 4.5: Atom probe sample preparation. (1) Trenches are milled to isolate a wedge-shaped section from the sample surface. (2) The sample is attached to the micromanipulator and lifted out. (3) The extracted lamella is positioned onto a micro-tip array, secured with Pt deposition, and then cut from the rest of it. (4) Final annular milling is performed to shape the tip into a needle with a diameter below 100 nm.

Liftout

The liftout process is performed under high vacuum conditions ($\sim 10^{-5}$ – 10^{-6} mbar). Throughout this process, a combination of FIB milling and SEM imaging is employed to achieve high precision and control at the microscale.

The process begins with sample mounting and grounding, using copper tape, which is essential to avoid charge buildup from the incoming ion and electron beams. The ROI is then located using SEM imaging at 0° tilt. To protect the ROI from beam-induced damage during milling, a protective layer of platinum (Pt) is deposited at 10 kV and 0.8 nA using a gas injection system (GIS).

Once protected, the sample stage is tilted to align the ion beam appropriately at 22° and an additional 300 nm Pt layer is deposited, with the FIB operating at voltage and current of 30 kV and 80 pA. The focused Ga^+ ion beam, operated at 30 kV and 2.5 nA, is then used to mill three sides of the desired volume, producing a wedge-shaped lamella with a triangular cross-section.

After defining the wedge, the stage is returned to 0° tilt to allow precise attachment of the micromanipulator to the free edge of the lamella. This attachment is secured by Pt deposition at the interface between the probe and the lamella. Following successful attachment, a final FIB cut is performed to release the lamella completely from the sample.

The extracted lamella is then transferred to a prefabricated microtip array, typically consisting of a 6×6 grid of silicon posts. At each site, a small segment of the lamella (~3 $\mu\text{m} \times 3 \mu\text{m}$) is brought into contact with a microtip and attached using Pt. A FIB cut is then made to separate the segment from the rest of the lamella. This process is repeated sequentially, positioning one wedge segment per microtip. After initial mounting, a second Pt weld may be applied from the opposite side of the lamella fragment to enhance mechanical stability during the subsequent sharpening process.

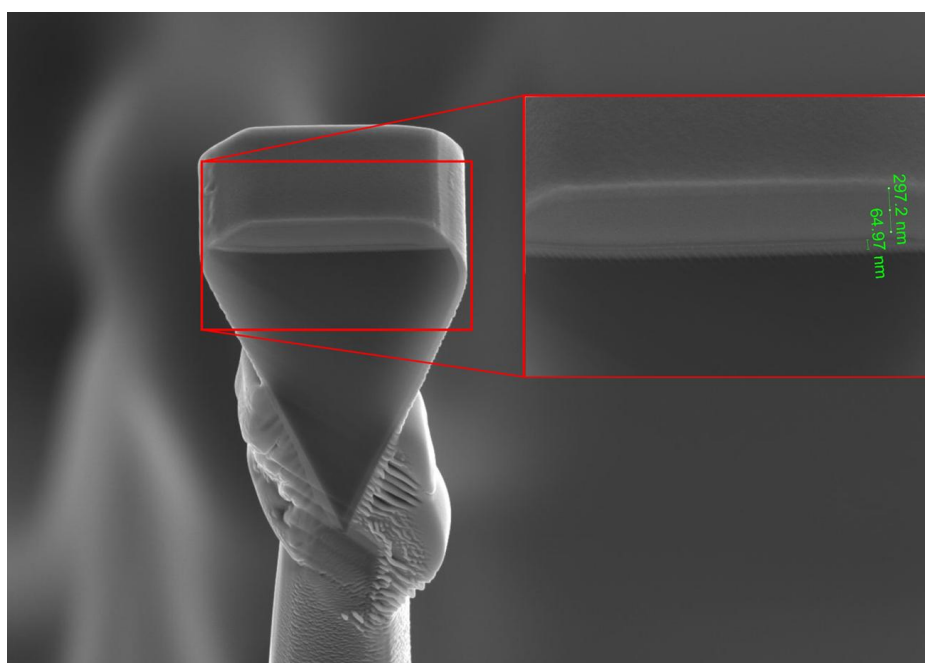


Figure 4.6: Cross-sectional view of the lamella mounted on a Si tip. The region of interest, the superlattice (SL), is visible as the bottom layer and measures approximately 60 nm in thickness. It is capped by a 300 nm thick protective Pt layer.

At the end of the liftout procedure, each microtip carries a small portion of the original ROI, precisely positioned and securely fixed. (Figure 4.6) These mounted tips are then ready for final shaping via annular milling to achieve the sharp, needle-like geometry required for successful atom probe tomography.

Tip Sharpening

Following the liftout process, the wedge-shaped lamella fragments attached to the microtip array must be sharpened into needle-like geometries suitable for atom probe tomography.

This is achieved through a FIB technique known as annular milling, performed under simultaneous SEM observation. The goal of this process is to produce tips with apex diameters typically below 100 nm, often approaching 50 nm, while preserving structural integrity and minimizing ion implantation.

To begin, the specimen stage is tilted to 52°, aligning the tip axis perpendicularly to the Ga⁺ ion column, a configuration necessary for concentric milling. An annular milling pattern is then applied using the FIB, operated at 16 kV, with beam currents adjusted according to the material properties and desired precision. For softer or more volatile materials, such as phase-change compounds, lower beam currents are preferred to reduce sputter damage and ensure better control over the milling.

The milling starts with shaping the lamella into a cylindrical form using an annular beam pattern with an outer diameter of approximately 4 µm and an initial inner diameter of 2 µm, which is gradually reduced, typically down to ~0.2 µm in a controlled, stepwise manner. Ensuring smooth transitions between the milling steps is critical, particularly in samples with compositional or mechanical contrasts, such as between a hard silicon substrate and a soft deposited layer, where abrupt shape changes can compromise the mechanical integrity of the tip.

Throughout this process, SEM imaging is used to monitor tip morphology and to examine the cross-sectional profile of the tip to ensure uniformity and detect embedded features.

Once the target geometry is achieved a final low-kV cleaning step is performed to remove surface contaminants, particularly gallium ions implanted during high-voltage milling. This is accomplished by irradiating the tip with a low-energy (5 kV) Ga⁺ beam at reduced current (e.g., 15 pA, 40pA) in a circular pattern. This shallow ion penetration, typically <10 nm, effectively removes the outermost Ga-contaminated layer without altering the tip geometry.

At the end of this procedure, the sample presents as a sharp, stable, and contamination-free tip, ready for atom probe tomography. The precision of the tip geometry, particularly the apex radius and shank angle, directly influences the quality of ion projection and the overall resolution of the atom probe dataset.

Tip Preparation for Site-Specific Atom Probe Analysis of Recrystallized Regions

In this work, a site-specific approach was employed to prepare APT tips centred on laser-induced recrystallized regions. The goal was to ensure that the apex of each tip encloses a defined recrystallized area for accurate analysis.

The original sample is a crystalline thin film grown by molecular beam epitaxy (MBE). To locally modify the structure, a laser pulse was first used to amorphize targeted regions of the sample. Immediately following this, a sequence of 20 laser pulses was applied to

recrystallize these areas. The resulting recrystallized spots were approximately 300 nm in size. However, these spots exhibited negligible contrast in scanning electron microscopy (SEM) relative to the surrounding crystalline matrix, making their precise location difficult to determine.

To overcome this challenge, a specific spatial pattern was employed (Figure 4.7). Sets of six recrystallized spots were written in a linear arrangement, each positioned between and aligned with intentionally destroyed or amorphized reference spots. These surrounding reference spots served as visual markers that were clearly distinguishable under SEM, enabling reliable localization of the otherwise indistinct recrystallized regions.

Initially, a protective platinum layer was deposited over the ROI containing the recrystallized features. Using the visible amorphized spots as guides, Pt marker lines were then deposited in the SEM to geometrically mark the positions of the recrystallized regions (in some cases, as shown in Figure 4.7, these markings were done via shallow cuts instead of Pt lines). The crossings of these Pt lines corresponded precisely to the intended recrystallized spots.

Following this, standard liftout procedures were carried out. Lamellae encompassing the marked recrystallized regions were extracted and mounted onto pre-sharpened Si posts. The final step involved annular milling to produce sharp APT tips, while maintaining control to ensure that the Pt line crossings remained centred at the apex of the tip.

This method allowed for high confidence in the spatial targeting of sub-micron recrystallized areas within the APT measurement volume, despite their lack of SEM contrast and nanoscale dimensions.

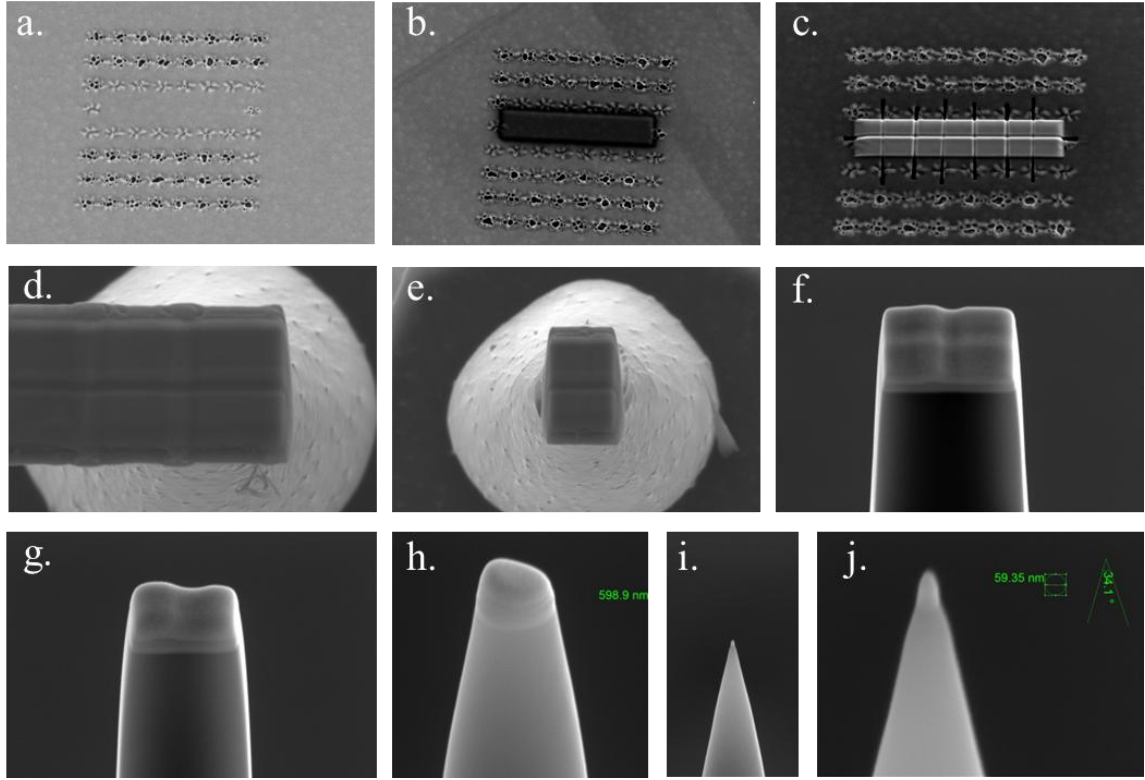


Figure 4.7: (a) Initial spatial arrangement of laser-written features: recrystallized spots aligned and bordered by amorphized reference spots. (b) Region after Pt deposition to protect the area of interest. (c) Marker lines added to locate the recrystallized spots, with intersections marking target areas. (d, e) Mounting of the lamella onto Si posts, with Pt line crossings centred. (f–j) Final tip sharpening process, maintaining the recrystallized region at the tip apex.

4.2 Laser Switching Setup (LSS)

For the optical switching of the SLs, a laser system was utilized as depicted in Figure 4.8. The system consists of a semiconductor pulsed laser diode driven by a PICOLAS module and control unit (PLCS-21), emitting at $\lambda = 660$ nm with a peak power of 400 mW. The PICOLAS driver transforms a set voltage U at the laser diode into an average pulse power

$$P = 37.89 \frac{\text{mW}}{\text{V}} \cdot U - 154.93 \text{ mW}.$$

The emitted beam is collimated through an aspheric lens ($\text{NA} = 0.5$) and subsequently directed towards a dichroic beam combiner via two silver mirrors. This combiner reflects most wavelengths between 630 and 690 nm while allowing others to pass through, guiding the beam into a $10\times$ objective ($\text{NA} = 0.25$), where it is focused onto the sample.

To facilitate alignment and visualization, a white LED source is introduced into the optical path via a beam splitter. The reflected light from the sample, including a small fraction of the laser signal, is collected through the objective and directed into a CCD camera after

passing through the optical elements. The camera provides real-time imaging to assist with precise targeting.

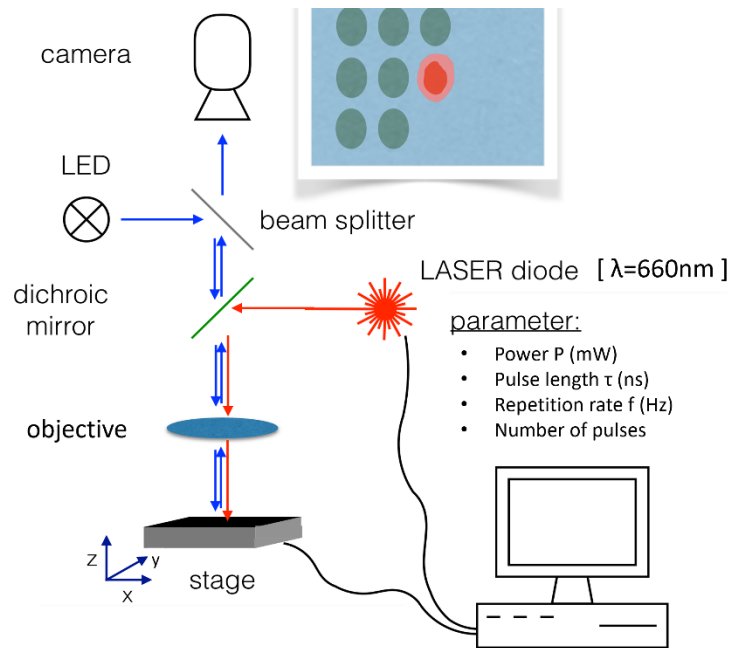


Figure 4.8: Schematic of the laser switching setup (LSS). A pulsed laser (at a wavelength of 660nm) with an elliptical Gaussian beam profile is emitted from the driver-controlled laser diode and is redirected onto the sample through a system of lenses and mirrors. [29, 31]

The sample is mounted on a high-precision NanoMax-TS stage (Thorlabs GmbH), which allows for movement in three dimensions. A closed-loop piezo controller enables automated positioning with nanometre-scale accuracy. A LabVIEW-based program integrates control over the piezo stage, LED illumination, and CCD camera, allowing for precise laser pulse delivery and live monitoring of the sample surface. [29, 31]

The laser pulses in this setup are defined by four key parameters: pulse duration (τ), pulse power (P), repetition rate (f) and number of pulses. The pulse duration can be adjusted within a range of 1 ns to 30 ns, with pulse power varying between 0 mW and 400 mW. The repetition rate is adjustable from 1 Hz to 2 MHz. Additionally, the sample stage, controlled via piezo actuators, allows for precise positioning with a range of 20 μm and step accuracy of 5 nm.

Due to the laser diode's active medium being much thinner than it is wide, the lateral Gaussian intensity distribution of the focused laser light at the sample surface takes on an elliptical shape. In the focus, a beam waist radius of 1.54 μm along the slow axis and 2.21 μm along the fast axis, was measured. [31]

4.3 Electron Backscatter Diffraction (EBSD)

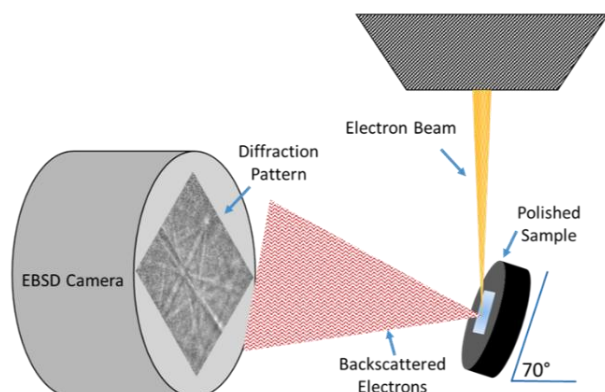


Figure 4.9: Schematic for EBSD analysis. A crystalline sample, tilted 70° from the normal beam incidence, generates a diffraction pattern from backscattered electrons, which is then captured on a phosphor screen camera. [44]

Electron Backscatter Diffraction (EBSD) is an SEM-based technique used to extract crystallographic information from a sample. In this method, a crystalline sample with a well-prepared surface is positioned inside the SEM chamber at a 70° tilt toward the EBSD detector. (Figure 4.9)

A stationary high-energy electron beam interacts with the tilted sample, producing a diffraction pattern composed of bright, straight Kikuchi bands. These patterns, captured by a CCD camera, follow the Bragg condition and contain essential details about the crystal structure and orientation at the specific measurement point. As a result, EBSD enables phase identification, crystallographic orientation mapping, and defect analysis, including grain boundaries and segregation. [45]

Since the diffracted electrons originate from only the top few nanometres of the sample, surface quality plays a crucial role in determining the clarity of the diffraction pattern.

4.4 Atomic Force Microscopy (AFM)

Atomic force microscopy (AFM) was used in this work to analyze the surface morphology of the samples at the nanoscale. AFM is a powerful technique capable of resolving surface topography with nanometer precision in the lateral (xy) direction and sub-nanometer (angstrom-level) resolution in the vertical (z) direction. It operates based on the interaction forces between a sharp tip and the surface potential of the sample, allowing it to be used on both conductive and non-conductive materials. In this study, measurements were carried out in tapping mode, where the cantilever is excited to oscillate near its resonant frequency.

As the tip scans over the sample surface, variations in surface height reduce the amplitude of the oscillation. A laser beam reflected off the back of the cantilever is used to detect these oscillation changes, which are then processed through a feedback loop to adjust the z-

position of the tip in real time. (Figure 4.10) From this feedback signal, the surface height profile is reconstructed. [46]

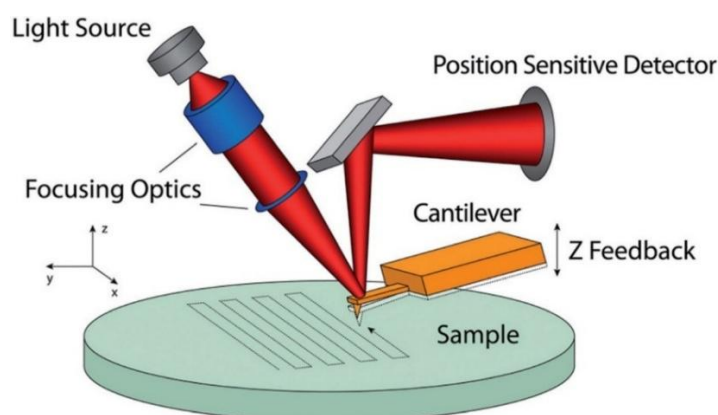


Figure 4.10: Schematic of the optical lever system in AFM, where a laser reflects off the cantilever onto a detector to track surface topography during scanning. [47]

4.5 Optical Microscopy

As mentioned previously, a distinct property of phase-change materials is the difference in optical reflectivity between their crystalline and amorphous states. The crystalline phase is highly reflective, whereas the amorphous phase exhibits much lower reflectivity and appears as darker regions within the crystalline matrix of the SL samples used in this work. This contrast allows us to effectively observe and measure the formation and dimensions of the amorphized areas using optical microscopy. For these measurements, a Keyence VHX-7000 microscope was utilized, operating at 6000 $\times$ magnification with a 200 nm resolution, ensuring detailed visualization of the targeted regions.

EXPERIMENTAL RESULTS

In this chapter, we present experimental results for the four GST-124/Sb₂Te₃ superlattice (SL) configurations. Our investigation spans the as deposited, amorphized, and recrystallized states, aiming to understand the mechanisms governing their switching behavior. We begin by examining the structural and chemical properties of the superlattices in their as-deposited state using atom probe tomography (APT). Optical switching experiments were then conducted to induce both amorphization and recrystallization, with the additional goal of examining whether the superlattices exhibit improved performance compared to common GST-124. Selected areas were subsequently characterized using techniques such as atomic force microscopy (AFM) and electron backscatter diffraction (EBSD). These combined measurements provide insight into the phase-change behavior and interfacial mechanisms within the SLs.

5.1 As deposited Superlattices

All atom probe tomography measurements presented in this thesis were performed using laser pulsing mode. The system utilized a pulsed ultraviolet (UV) laser with a wavelength of 355 nm and a pulse duration of 10 ps. For each measurement, the pulse repetition rate was set to either 200 or 250 kHz, and maintained a base temperature of 30 or 40 K. These parameters were fixed during each individual experiment but could be adjusted between measurements to optimize data quality. A constant detection rate of 0.5% was applied throughout. The laser pulse energy ranged from 16 to 21 pJ depending on the sample and conditions. For every sample studied, at least three successful APT measurements were conducted for statistical purposes.

Structure of As-Deposited Superlattices derived by APT

The structural and chemical characteristics of the as deposited superlattices are examined to establish a well-defined reference prior to switching. Atom probe tomography provides nanoscale insights into the composition and layering quality, serving as a basis for comparison with the amorphized and recrystallized states. Figure 5.1 shows representative 3D reconstructions obtained from atom probe tomography for each of the superlattice samples. The GST-124 layers are visualized in red, corresponding to the presence of Ge atoms. To improve the visibility of the layer structure, an adjusted isosurface of Ge at. % is displayed. The Sb_2Te_3 layers appear in green, as they are dominated by Te, clearly illustrating the intended layered structure. Note that the thickness of each superlattice in the reconstruction was calibrated to correspond to the XRD calculated thickness when the whole superlattice is measured from the capping (In or Pt) to the Si substrate or adjusted to the number of visible layers. For the superlattices with larger supercell thickness, N:8, N:6, and N:4, the number and spacing of the layers align well with the designed architecture. In the case of the N:2 sample, however, individual layers are more difficult to resolve. This is expected, as the supercell thickness is approximately 4.5 nm, approaching the spatial resolution limit of atom probe tomography.

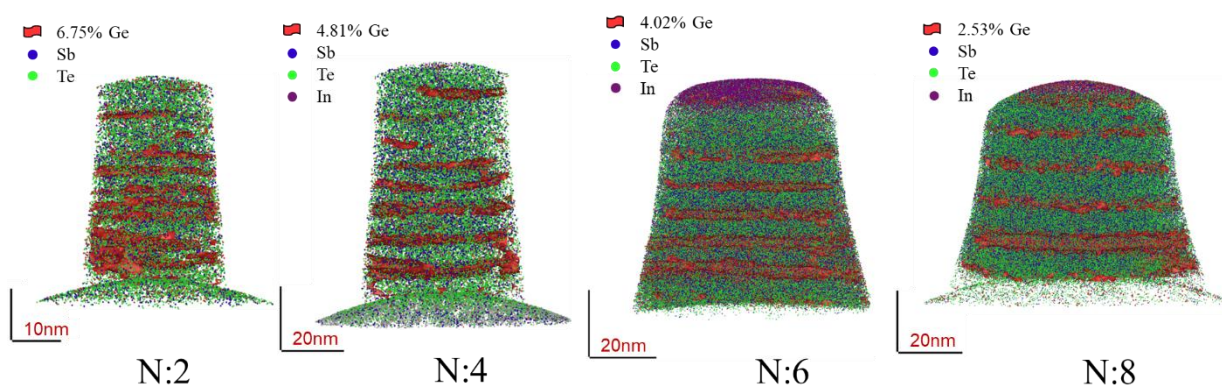


Figure 5.1: 3D reconstructions of the superlattices from left to right in order of increasing supercell thickness.

In addition to the 3D reconstructions, 1D concentration profiles were extracted from each successful atom probe measurement. (Figure 5.2) Measurements were considered successful if at least half of the expected number of layers could be identified. In addition, the background level and the correlation histogram were always taken into account when evaluating measurement quality.

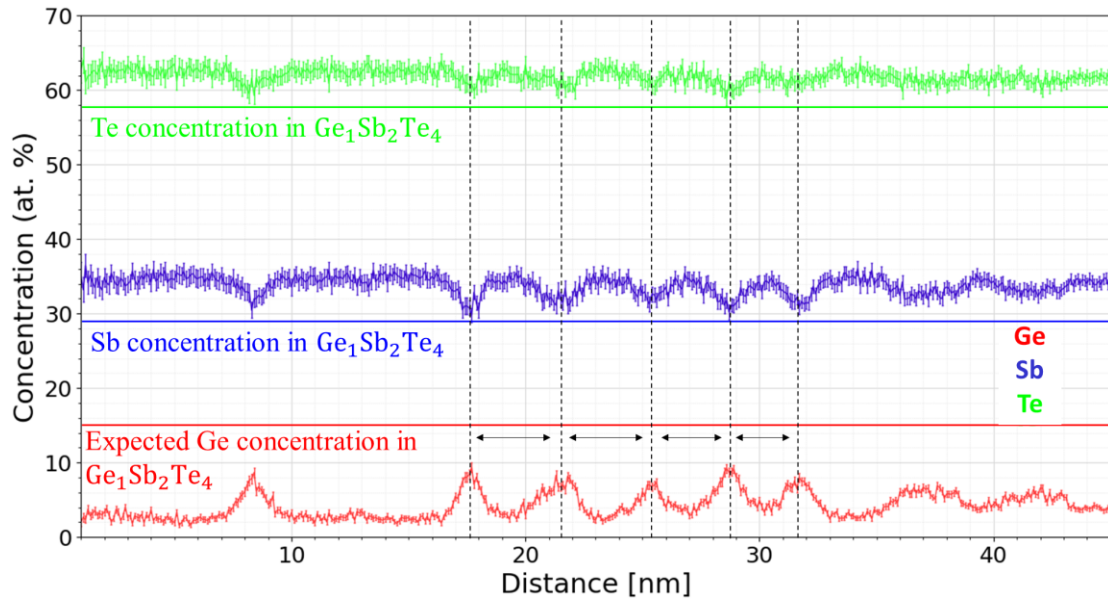


Figure 5.2: Concentration profile for the N:2 superlattice. Intermixing between GST-124 and Sb_2Te_3 is clearly noticed by the presence of ~3 at. % Ge in Sb_2Te_3 . Horizontal lines indicate the expected concentrations of Ge, Sb and Te for the GST-124 configuration. Vertical lines mark consecutive Ge peaks, with the spacing between them corresponding to the supercell thickness.

In these profiles, the periodic peaks in the Ge concentration signal correspond to the GeSb_2Te_4 layers, allowing clear identification of the multilayer structure. The distances between successive GST layers were then extracted from the profiles and used to calculate the supercell thickness Λ_{SL} for each tip. An average supercell thickness was determined for each configuration and was compared to values obtained from X-ray diffraction (XRD), as shown in Figure 5.3. All deviations from the desired period thicknesses in the samples are smaller than 0.6 nm, corresponding to relative deviations of less than 10%. Overall, this indicates good agreement between the two techniques. For the thinner superlattices, N:2, N:4, and N:6, APT yields slightly larger values than XRD, which is consistent with known limitations of the method. Factors such as local magnification effects, where differences in evaporation fields between materials can distort the apparent structure, trajectory aberrations, limited spatial resolution, and intermixing due to laser-induced heating can contribute to apparent layer thickening in the reconstructions [41]. Nonetheless, the measurements confirm that the intended layering is preserved in the as-deposited samples and validate the use of APT for structural characterization at the nanoscale, despite operating near the resolution limits of the technique.

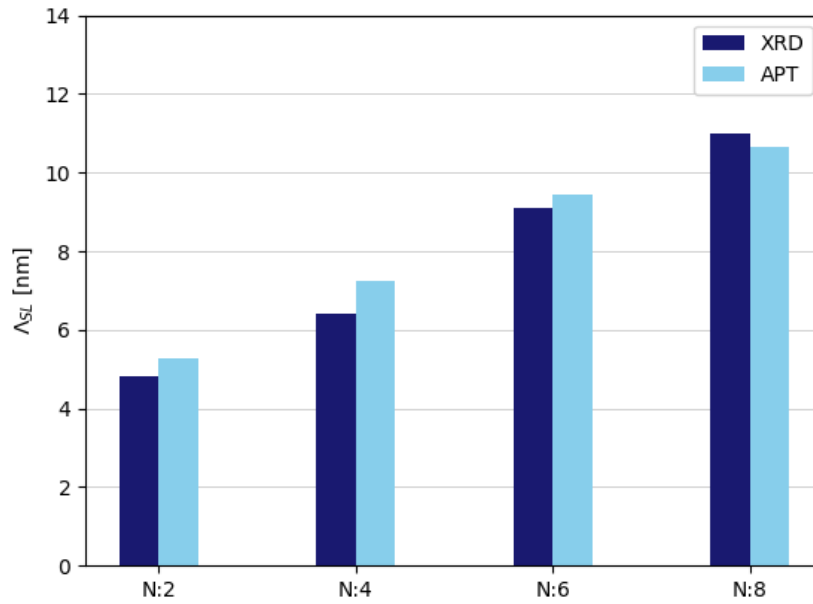


Figure 5.3: Comparison of APT- and XRD-derived supercell thickness Λ_{SL} .

Chemical Composition of as deposited Superlattices derived by APT

In addition to structural analysis, atom probe tomography provides valuable insight into the chemical composition of GST-124 / Sb_2Te_3 superlattices. By extracting elemental concentration profiles across the layers (Figure 5.4), we observe clear deviations from the nominal stoichiometries of GST-124 and Sb_2Te_3 . These deviations indicate intermixing, already visible in the thinnest supercell superlattice (N:2), where approximately 3 at. % Ge is detected in the Sb_2Te_3 layers, despite Ge not being a constituent of this compound.

To quantify the composition of both GST-124 and Sb_2Te_3 layers, elemental concentrations were extracted from the 1D profiles at the positions of maximum Ge content (for GST-124) and maximum Sb content (for Sb_2Te_3). These values were then averaged across all superlattices to provide an average representative composition. The results are summarized in Table 3. To facilitate comparison with established GST compounds, the stoichiometries are expressed in the form $\text{Ge}_x\text{Sb}_y\text{Te}_4$ for GST-124 and $\text{Ge}_x\text{Sb}_y\text{Te}_3$ for Sb_2Te_3 , acknowledging that Te remains relatively immobile, while Ge and Sb are more prone to interdiffusion.

The chemical composition extracted by APT reveals deviations from the ideal stoichiometries of both GST-124 and Sb_2Te_3 , indicating significant elemental intermixing between the layers.

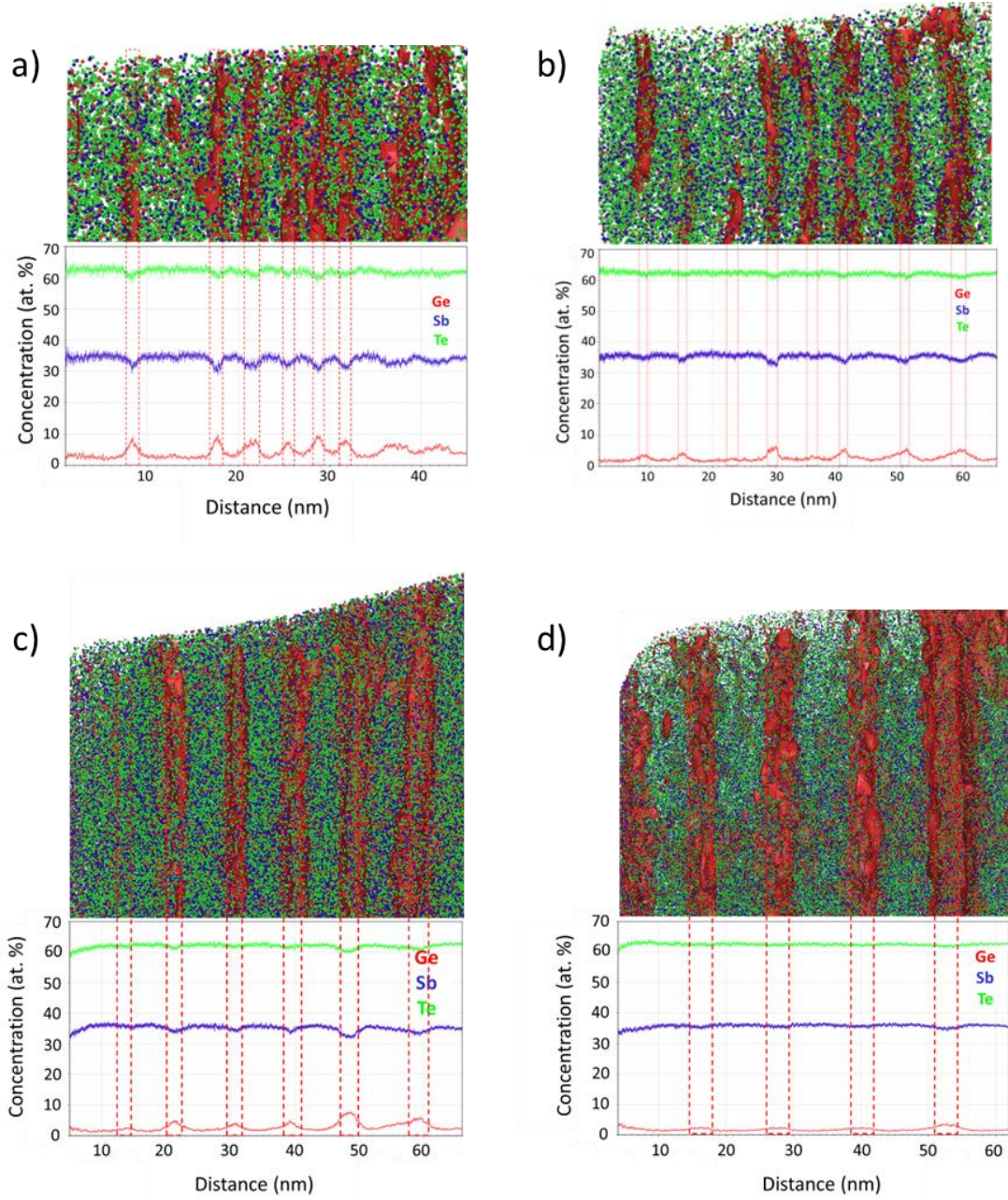


Figure 5.4: APT investigation of the PCM SLs. a) to d) the PCM SLs in order of increasing Sb_2Te_3 layer thickness. 3D elemental maps each with the corresponding 1D concentration profile of Ge, Sb and Te across the GST- 124 / Sb_2Te_3 superlattices. Intermixing between GST- 124 and Sb_2Te_3 is noticed by the presence of $\sim 2\text{--}3$ at. % Ge in Sb_2Te_3 .

A clear trend that emerges from the comparison is that as the superlattice period increases, from N:2 to N:8, the Ge content within the GST layers systematically decreases (from ~ 6.5 at. % to ~ 3.3 at. %), while its presence in Sb_2Te_3 remains low but non-negligible ($1.6\text{--}3.2$ at. %), highlighting the asymmetry in interdiffusion behavior. This suggests enhanced intermixing, likely due to Ge atoms diffusing more extensively into the increasingly thicker

Sb₂Te₃ layers. In thicker supercells, Ge atoms must travel farther to reach the next GST region, thus increasing the probability of incorporation into neighboring Sb₂Te₃.

Region	Ge (at.)	Sb (at.)	Te (at.)	Stoichiometry
GST- 124	14.29	28.57	57.14	Ge ₁ Sb ₂ Te ₄
Sb ₂ Te ₃	0	40	60	Sb ₂ Te ₃

Table 2: Expected stoichiometry o GST-124 and Sb₂Te₃.

In contrast, Sb shows a weaker but noticeable trend, with slightly elevated concentrations in the GST layers and corresponding depletion in Sb₂Te₃, compared to their ideal values. This suggests some degree of Sb diffusion in the opposite direction, albeit less pronounced than that of Ge. Te remains the most stable element across all samples, showing only minimal variation between layers and stack periods. This stability justifies its use as a normalization reference for stoichiometric comparisons.

Sample	Region	Ge (at.)	Sb (at.)	Te (at.)	Stoichiometry
N:2	GST- 124	6.52 ± 1.14	32.34 ± 1.04	60.73 ± 0.53	Ge _{0.43 ± 0.08} Sb _{2.13 ± 0.07} Te _{4 ± 0.04}
	Sb ₂ Te ₃	3.23 ± 1.07	34.81 ± 0.83	62.22 ± 0.46	Ge _{0.16 ± 0.05} Sb _{1.68 ± 0.04} Te _{3 ± 0.02}
N:4	GST- 124	4.47 ± 0.92	33.74 ± 0.53	61 ± 0.39	Ge _{0.29 ± 0.06} Sb _{2.23 ± 0.04} Te _{4 ± 0.03}
	Sb ₂ Te ₃	2.14 ± 0.34	35.71 ± 0.43	62.39 ± 0.19	Ge _{0.1 ± 0.02} Sb _{1.72 ± 0.02} Te _{3 ± 0.02}
N:6	GST- 124	4.07 ± 1.6	34.16 ± 0.86	61.51 ± 0.66	Ge _{0.26 ± 0.1} Sb _{2.22 ± 0.06} Te _{4 ± 0.04}
	Sb ₂ Te ₃	1.63 ± 0.25	35.75 ± 0.22	62.76 ± 0.27	Ge _{0.08 ± 0.01} Sb _{1.71 ± 0.01} Te _{3 ± 0.01}
N:8	GST- 124	3.27 ± 0.84	34.99 ± 1.99	62 ± 1.06	Ge _{0.2 ± 0.05} Sb _{2.17 ± 0.12} Te _{4 ± 0.07}
	Sb ₂ Te ₃	1.79 ± 0.5	34.37 ± 2.13	63.98 ± 1.51	Ge _{0.08 ± 0.02} Sb _{1.61 ± 0.1} Te _{3 ± 0.07}
GST-124		20.12 ± 2.77	26.71 ± 1.93	54.25 ± 1.57	Ge _{1.5 ± 0.14} Sb _{1.98 ± 0.11} Te _{4 ± 0.06}

Table 3: Elemental concentrations of the four superlattice configurations and GST-124 reference sample. The concentration values are averaged over all APT measurements for each sample. The numbers given are mean elemental concentrations and standard deviations.

Overall, the data points to a non-uniform intermixing mechanism, where Ge diffuses more readily into Sb_2Te_3 than Sb into GST. As the superlattice period increases, interlayer diffusion becomes more prominent, leading to a gradual transformation of the intended discrete layers into regions with metastable compositions. APT's inability to resolve atomic stacking sequences limits definitive structural assignments. Therefore, while the stoichiometries suggest the possible incorporation of Sb within GST and Ge into Sb_2Te_3 , high-resolution TEM would be required to directly verify the formation of partially intermixed, compositionally modulated phases indicated by APT.

2D Defects in Superlattices

Up to this point, the analysis has centered on the overall properties of the superlattices (SLs), treating them as uniform, defect-free materials. The following section shifts focus to localized features, specifically, two-dimensional (2D) defects within the SLs. Moving beyond the investigation of idealized bulk structures, atom probe tomography (APT) reveals a range of compositional inhomogeneities and structural deviations in the crystalline superlattices across all studied configurations. These nanoscale defects can be broadly categorized into conventional defects, such as grain boundaries, and more unconventional 2D features, including variations in layer thickness, irregular stacking, and bridging structures that form between layers. [48]

Unconventional 2D Defects

A range of stacking-related anomalies were observed in the APT measurements of crystalline SLs. These include:

- Variations in layer thickness, where individual layers are significantly thinner or thicker than the nominal periodicity. Such deviations may be interpreted as stacking faults or interruptions in the regular growth of the superlattice (Figure 5.5 d).
- Glued or merged layers, particularly in regions where two adjacent layers of the same type, most commonly GST-124, appear directly connected, lacking the expected intervening phase (Figure 5.5 c).
- Dissolved or incomplete layers, where the typically well-defined GST-124 disk-shaped layers appear discontinuous or significantly diminished in the APT data, indicated by a reduced or missing Ge signal (Figure 5.5 a).

Although these features disrupt the ideal periodicity of the SLs, their elemental compositions are generally similar to the surrounding material. This suggests that they do not represent the formation of entirely new phases but rather reflect localized growth instabilities or kinetic limitations during the deposition process.

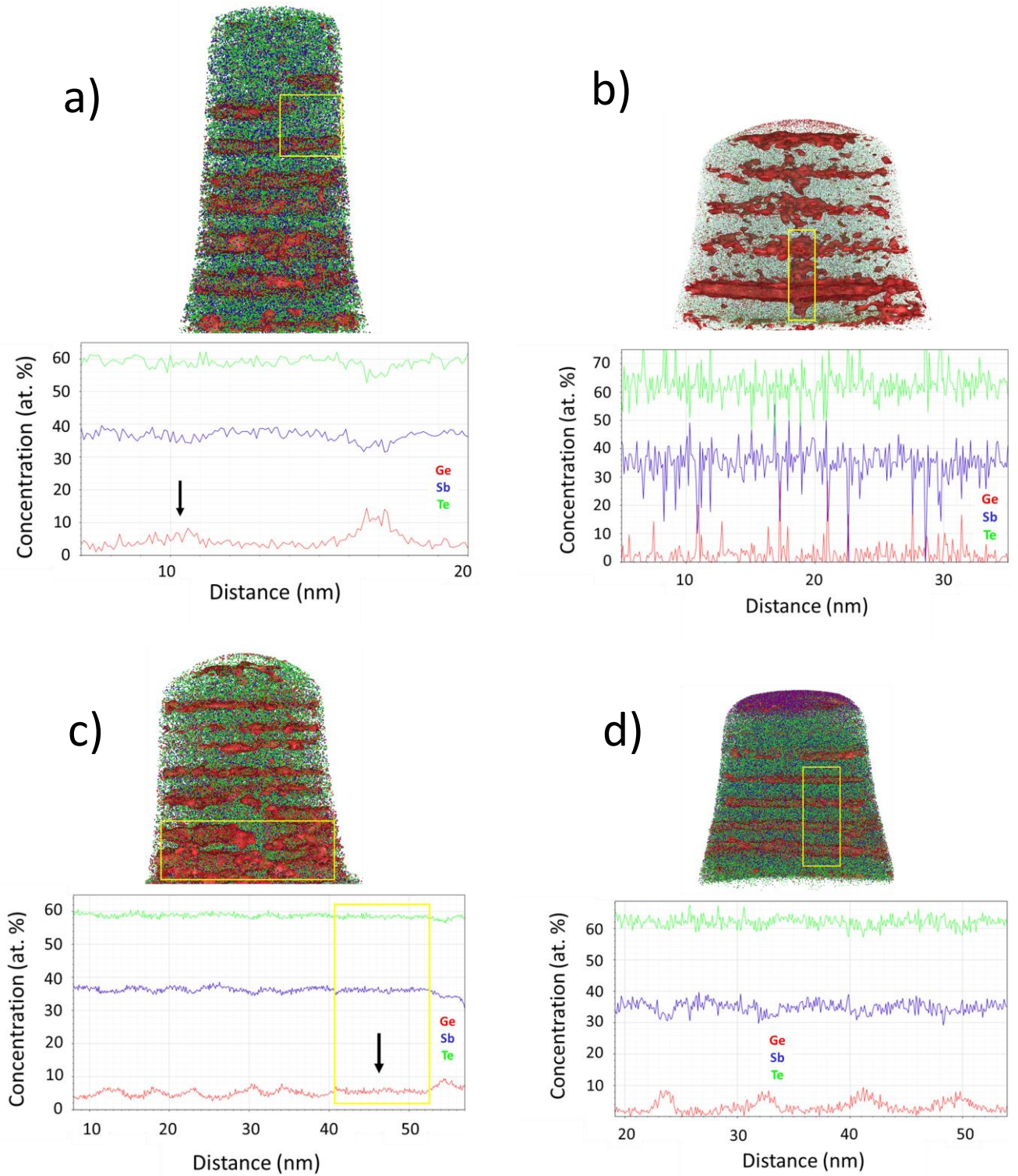


Figure 5.5: Unconventional 2D defects observed in APT measured as deposited crystalline SLs, highlighted with yellow rectangles. a) dissolved GST-124 layers appear as lower or disappeared peak of Ge concentration, b) 1D concentration profile of Ge, Sb and Te across a Ge bridge, c) 1D concentration profiles across region of “glued” GST-124 layers. The position of the defect is indicated by black arrow. d) Different layer thickness, the first two layers in the yellow box have ~ 1.7 nm thickness, while the last two are ~ 3.3 nm thick.

Another frequently observed 2D defect is the formation of narrow, bridge-like connections between adjacent layers of similar composition. These structures, often linking neighboring GeSb_2Te_4 or Sb_2Te_3 layers, may arise due to enhanced atomic mobility or localized diffusion during growth and represent a significant deviation from the intended layered architecture (Figure 5.5 b).

The compositional and structural analysis of these nanoscale features highlights that intermixing, irregular stacking, and unconventional 2D defects are intrinsic to the as-deposited state of the superlattices. Defects such as stacking irregularities, glued layers, and atomic bridges represent key structural motifs that may influence the electrical, thermal, and phase-change behavior of the material. Understanding the origin, distribution, and nature of these defects is therefore essential for optimizing the functional performance of phase-change superlattices.

5.2 Optical Switching

Phase-change materials (PCMs) are gaining significant interest for photonic computing applications due to their potential for fast, energy-efficient, and multi-level optical switching. Unlike conventional binary electronic memories, PCMs can exhibit intermediate crystallization states, enabling analog control of optical properties essential for programming synaptic weights in optical neural networks and enhancing data storage density and processing speed.[49-51] Therefore, optical switching presents an approach to switching that offers a multitude of compelling capabilities and possibilities. To our knowledge, such optical switching has not yet been demonstrated for the PCM superlattice systems, underscoring the novelty of the present results.

5.2.1 Amorphization

For the optical switching experiments, each sample was ex-situ capped with a 25 nm layer of $\text{ZnS}:\text{SiO}_2$. This capping layer enhances the absorption of the laser, helping to localize the energy in the phase-change material (PCM) and making the switching process more efficient. To amorphize the material, a single laser pulses with a high power and a short pulse duration was used. That way the risk of recrystallization is avoided, which can occur when multiple pulses are applied.

To systematically study the effect of different laser parameters, the Laser Switching Setup (LSS) was employed to apply pulses in a 6×6 grid, where each position corresponds to a unique combination of pulse power and pulse duration. In total nine 6×6 grids were performed and then combined to build a larger map covering a broad range of laser settings.

In the resulting optical microscope images, the amorphous regions appear as darker spots compared to the surrounding crystalline matrix due to their lower reflectivity (Figure 5.6 a). The area of each amorphized spot was measured using a built-in feature in the optical microscope, which automatically calculates the size of a selected region. The final values used for analysis are the average areas obtained from the three repetitions of each grid.

This data is visualized in Power Time Effect diagrams (Figure 5.6 b), where the laser power and pulse duration are plotted in the y and x axes respectively. The color scale indicates the average area of the amorphized spot. These diagrams provide a clear overview of how the switching behavior depends on the laser parameters, helping to identify amorphization thresholds, growth trends, and optimal switching conditions.

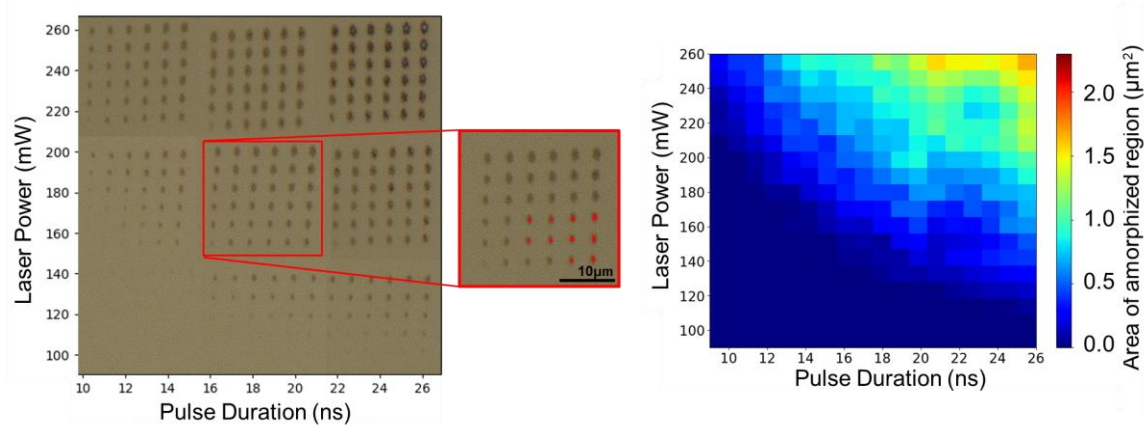


Figure 5.6: a) Grid of nine 6 x 6 points performed in the LSS for the as deposited N:8 SL. The parameters of the laser pulse applied are 90-260 mW and 9-26 ns. Each point corresponds to an amorphous spot where the applied laser amorphized the sample. Inset: Red areas are the evaluated amorphized spot sizes, as determined by the software of the optical microscope. b) Power-Time-Effect (PTE) diagram. The length and the power of the laser pulse are represented in the x and y axis respectively, and the amorphized spots' area in the z axis in color-code. The sample was crystalline as deposited, and for short laser pulses with low power, the material remains crystalline, whereas stronger laser parameters result in successful amorphization.

The Power-Time-Effect (PTE) diagrams obtained for all the samples, corresponding to both the first and second amorphization cycles, are shown in Figure 5.7. As expected, regions in the lower left corner of each PTE diagram, representing low pulse power and short pulse duration, show zero amorphized spot area, indicating that the laser pulse is insufficient to induce a phase change. In these regions, the material remains in its initial crystalline state.

However, as both laser power and duration increase, the applied energy becomes sufficient to amorphize the material, and the amorphized spot area increases, reaching values up to approximately 2 μm².

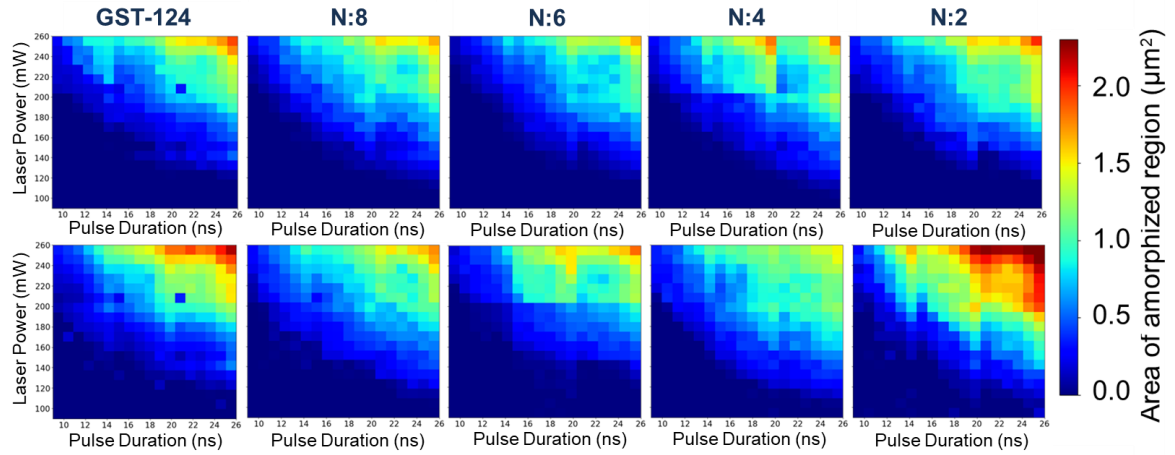


Figure 5.7: Amorphization PTEs for all the samples, for the first (top) and second (bottom) amorphization cycle. Each PTE resulted from single laser pulses of varying power 90-260 mW and duration 9-26 ns.

While the full PTE diagrams provide a comprehensive overview of the switching behavior, direct visual comparisons between different samples can be difficult. To simplify the comparison, the average amorphized spot area along the diagonal of each PTE diagram was extracted and plotted. (Figure 5.9) This diagonal corresponds to a range where switching occurs reliably without reaching regimes of excessive energy input. It is important to note that in some samples, particularly in the upper right corner of the PTE diagrams, there is a sudden increase in spot size, likely due to ablation or damage caused by overly strong laser pulses. This damage is also observable with light microscopy and AFM, and these data points are excluded from the comparative analysis.

By comparing the diagonal region, we are able to assess the relative switching performance of the samples. This approach reveals several notable trends, firstly the Sb_2Te_3 sample shows the smallest amorphized areas, consistent with its known limitations as a phase-change material compared to alloys like GST. The GST-124 sample shows higher amorphization performance than Sb_2Te_3 but generally lower than that of the superlattice structures, with the exception of the N:6 superlattice. (Figure 5.9) Finally, among the superlattice samples, a clear trend is observed: thinner supercell thicknesses lead to larger amorphized spot sizes, suggesting enhanced switching efficiency. This is further supported by Figure 5.8, showing the area of amorphized region plotted against total film thickness of each sample, which reveals no clear correlation, indicating that the observed effect arises from the superlattice structure itself rather than differences in overall film thickness.

This comparative analysis highlights the advantages of certain superlattice designs in achieving more effective amorphization under the same laser conditions, providing valuable insight for optimizing PCM structures for optical switching applications.

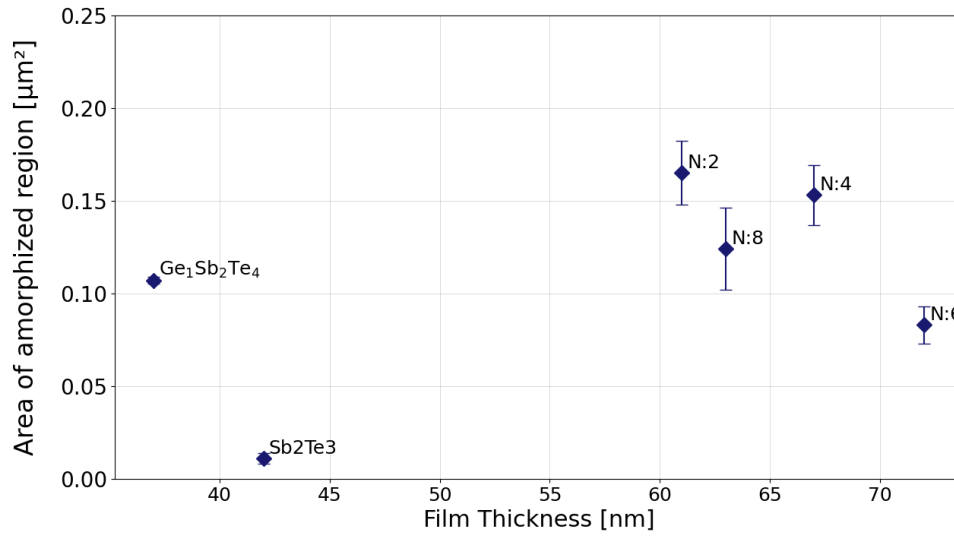


Figure 5.8: Amorphized region size as a function of total film thickness for all superlattice and reference samples. The absence of a clear correlation indicates that the variation in switching behavior is not due to differences in overall film thickness.

A second amorphization cycle was also performed, which corresponds to an amorphization – recrystallization – reamorphization cycle. In this second amorphization cycle, the overall trends observed in the first cycle persist. (Figure 5.9) All samples exhibit consistently larger amorphized areas compared to the first cycle, including the N:6 superlattice, which continues to show improved performance. This behavior mirrors what is often observed in electrical switching experiments, where several initial cycles are typically required before the material stabilizes into repeatable resistance states. [52] While the improvement seen here is less pronounced than in electrical switching, it is still noticeable. This difference can be attributed to the more efficient energy delivery and reduced thermal dissipation in electrical switching. Additionally, electrical switching is typically confined to nanometre-scale areas [7], whereas micrometre-sized areas are switched in our experiments. This difference in scale, combined with the known reduction in switching energy and reset current for smaller bottom electrodes in PCM cells [7, 12], may also contribute to the observed behavior and should not be ruled out.

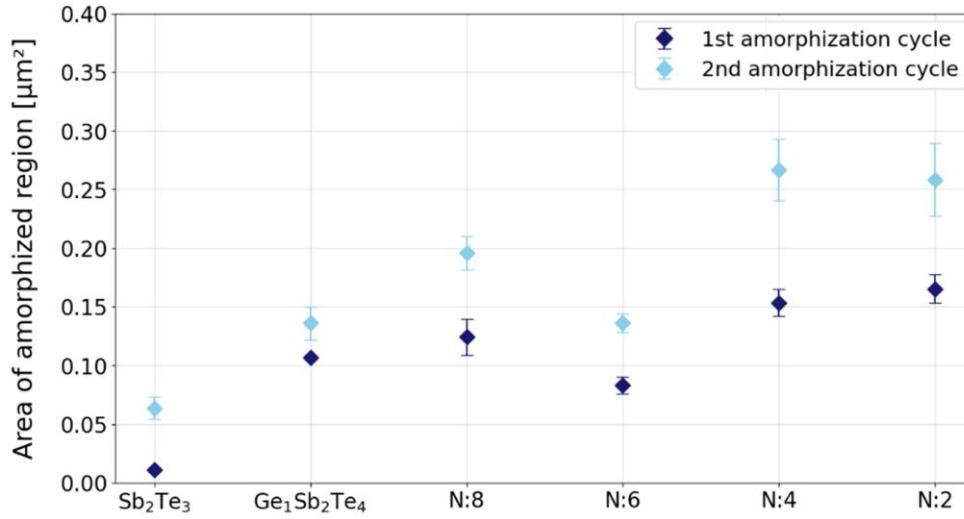


Figure 5.9: Comparison of the area of the amorphized region for the first and second amorphization cycle over all the samples.

The laser parameters for the optical switching, including amorphization and recrystallization parameters can be found in more detail in the APPENDIX, Table 5.

Correlative Analysis of Amorphized Regions - Optical microscopy, AFM, EBSD, APT

To enable detailed structural and compositional analysis of the amorphized phase, as well as to define a clean starting point for the subsequent recrystallization, a reliably amorphized region has to be selected. The N:8 superlattice was selected as the focus and to validate the phase state and quality of the selected region, a correlative analysis was performed using optical microscopy, atomic force microscopy (AFM), and electron backscatter diffraction (EBSD). The goal was to identify a set of laser parameters that produces a well-defined, uniform amorphous region, suitable for high-resolution techniques such as atom probe tomography.

AFM measurements of the candidate amorphized regions revealed clear differences in topography depending on the applied laser parameters. (Figure 5.10) In the selected amorphized spot for the profile line 1 in Figure 5.10, produced under moderate and well-controlled laser conditions, a height increase of approximately 10–12% was measured. This aligns well with the expected range for amorphous transitions in Ge-Sb-Te-based phase change materials, which typically exhibit a volume expansion of about 6–10%. [53, 54] The observed increase confirms that the spot resulting for laser power 170 mW and pulse duration 16 ns is structurally intact and reliably amorphized, averaging approximately $0.3 \mu\text{m}^2$ in size, making it suitable for further APT measurements.

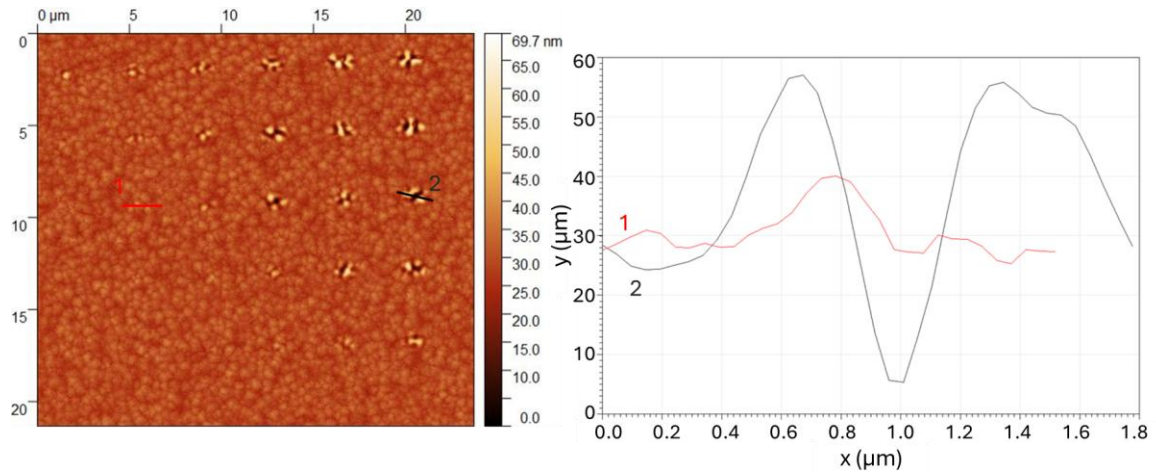


Figure 5.10 AFM scan after HF-dip to remove ZnS:SiO₂-capping and line profiles from two amorphous spots. Profile line 1: P = 170 mW, t = 16 ns and profile line 2: P = 170 mW, t = 20 ns

In contrast, spots formed using stronger laser pulses, like the one for the profile line 2 in Figure 5.10, are typically located toward the upper-right corner of the PTE diagrams and displayed pronounced surface irregularities. These regions showed hills and valleys with height variations of about 30 nm, forming fingerprint-like structures. These features usually originate from the ZnS:SiO₂ capping layer, however, since the AFM scans were performed after the capping layer was removed via HF etching, the exact implications behind these surface features remains unclear. These irregularities suggest structural damage, possibly due to partial ablation or interference-driven energy localization during switching. Furthermore, the non-periodic morphology that was observe is more consistent with thermal effects and mechanical deformation, potentially amplified by the Gaussian beam profile and the strong absorption enhancement from the ZnS:SiO₂ capping layer (prior to removal).

Previous studies have shown that such ridge-like features, persist and likely indicate permanent disruption of the layer stack, including cracking of the capping layer. In contrast, samples with a thicker capping layer show a much more stable response. The additional thickness appears to buffer mechanical stresses and reduce topographic artifacts, while also improving thermal efficiency during switching. [55]

Additionally, EBSD was performed in the same region to confirm amorphization. (Figure 5.11) The EBSD scan is presented as an Inverse Pole Figure (IPF) map overlaid with the Confidence Index (CI), which indicates the reliability of orientation measurements. In the EBSD map, the red regions indicate the out-of-plane orientation of the as-deposited crystalline matrix. In contrast to the crystalline matrix, amorphous regions appear as pure black spots in the EBSD map due to the absence of Kikuchi patterns and the resulting low Confidence Index (CI) values. These areas do not yield meaningful orientation data, and while the raw orientation may vary arbitrarily from point to point, such variations are not physically meaningful due to the absence of long- or short-range crystalline order.

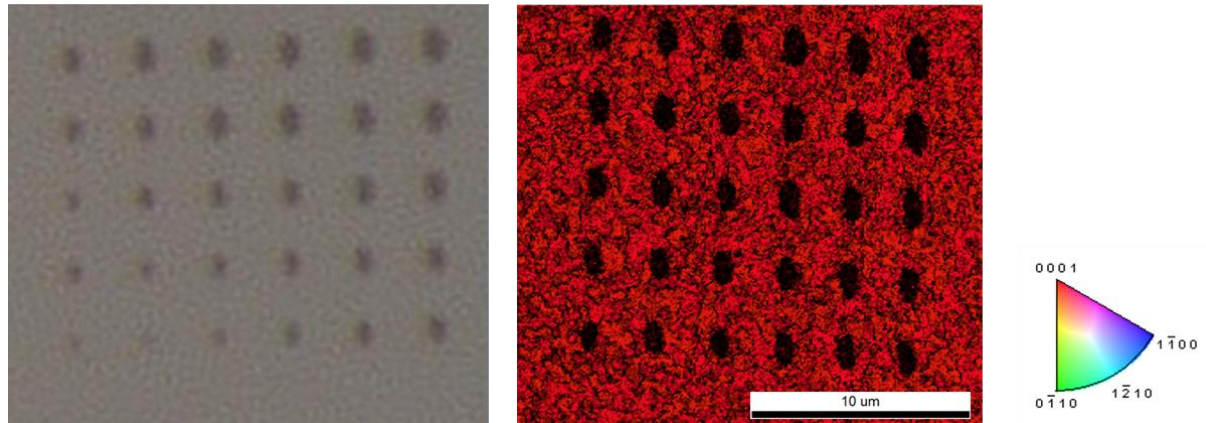


Figure 5.11 Optical microscope image for the amorphization of the N:8 superlattice and the corresponding EBSD scans (IPF with CI). The red surrounding area is the 0001 out of plane orientation of the as deposited crystalline matrix and the black spots correspond to the amorphous spots.

The CI overlay effectively suppresses this unreliable data, preventing misinterpretation. For spatial reference, an optical microscope image of the same area is included, in which the laser pulse locations align precisely with the dark amorphous spots in the EBSD map, confirming successful switching and localized amorphization.

Therefore, for further characterization in APT and reliable recrystallization studies, regions formed using laser pulses of 170 mW power and 16 ns duration, exhibiting strong optical contrast, volume change within expected limits ($\sim 10\text{--}12\%$), as was concluded by the AFM analysis, and a clear amorphous signature in EBSD, were selected as reliably amorphized areas.

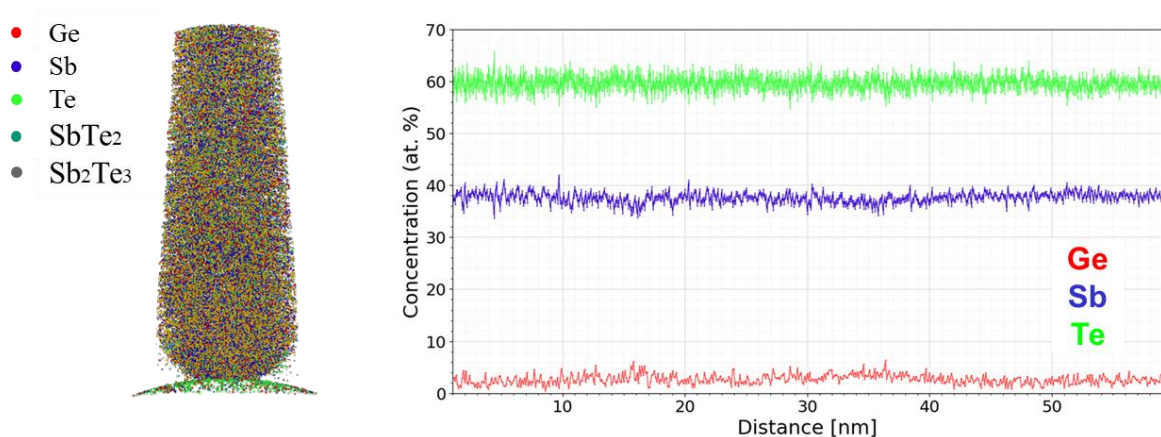


Figure 5.12 Amorphization area for laser power 170 mW and pulse duration 16 ns for the N:8 superlattice measured in APT. The periodic structure is lost and the Ge, Sb and Te concentrations are constant along the tip.

To investigate the structural and compositional state of the amorphized region at the atomic scale, atom probe tomography was performed on the amorphized spots using the selected laser parameters (170 mW, 16 ns). The resulting APT reconstruction and corresponding concentration profiles reveal a complete loss of the superlattice layering. (Figure 5.12) In the amorphized state, the elemental distribution is uniform, with measured concentrations of Ge $\approx$ 4%, Sb $\approx$ 37%, and Te $\approx$ 59%. This composition closely resembles that of Sb₂Te₃, nominally 40% Sb and 60% Te, with only a minor contribution from Ge. This homogeneous composition further confirms the collapse of the periodic structure characteristic of the superlattice crystalline phase.

Additionally, a clear reduction in the average overall PME is observed, dropping from 71% in the as deposited crystalline N:8 superlattice to 47% in the amorphous state. While this decrease is expected upon amorphization, it is noteworthy that the PME remains relatively high compared to typical values for covalently bonded phase-change materials, which generally exhibit PME values around 20%. One possible explanation could be that some degree of short-range ordering or clustering persists even in the amorphous state, where covalent bonding would be anticipated, potentially influenced by the material's superlattice origin and its surrounding matrix. Alternatively, the elevated PME might also stem from limitations or artifacts related to APT measurements on amorphous PCMs, such as laser-induced recrystallization. This ambiguity could not be resolved within the scope of this work and warrants further investigation.

5.2.2 Recrystallization

The next step in the switching cycle is the recrystallization of the previously amorphized regions. Starting from the as-deposited crystalline state, well-controlled single laser pulses of 170mW and 16ns were first applied to achieve amorphization. The next objective is to reverse this process and induce recrystallization. To identify the optimal conditions for recrystallization, not only the laser power and pulse duration but also the number of pulses applied were systematically varied, with 20 laser pulses resulting in the most reliable recrystallization. The laser parameters used in this study ranged from 0 to 70 mW in power and from 500 to 2000 ns in pulse duration.

GeSb₂Te₄

The recrystallization analysis started with the GST-124 reference sample. The optical microscope image of the laser-induced recrystallization process alongside the corresponding PTE diagram are presented in Figure 5.13. During amorphization, the as-deposited sample appears bright and optically uniform, consistent with its crystalline state. Upon laser exposure, localized amorphous regions are created, which appear as distinct dark spots under the optical microscope. In contrast, the recrystallization process begins from this state, with isolated, dark amorphized spots embedded in the otherwise crystalline

and bright matrix. As laser pulses of varying power and pulse duration intended for recrystallization are applied and focused on these dark regions, they gradually reduce in size and intensity. A progress of the recrystallization is indicated by a fading or re-brightening of the spots, as they begin to visually resemble the surrounding crystalline material. This optical progression serves as a clear indicator of the material's response to the applied laser parameters. For a successful and complete recrystallization, these initially dark, amorphized regions become optically indistinguishable from the surrounding crystalline matrix.

The optical microscope image of the GST-124 sample shows a clear evolution of the switched spots across the laser parameter space, as described in the previous paragraph. In the lower-left region, where laser power and pulse duration are low, the spots remain dark, indicating that recrystallization has not occurred. Moving toward the upper-right, the spots gradually fade and brighten, blending into the surrounding crystalline matrix. This visual recovery indicates successful recrystallization. Additionally, at the highest laser powers in the top row, some spots show damage or ablation, appearing as dark regions darker than the amorphous spots, often with a lighter central area. These observations are reflected in the accompanying PTE diagram, where the colour progression captures the correlation between laser parameters and the sample's recrystallization response.

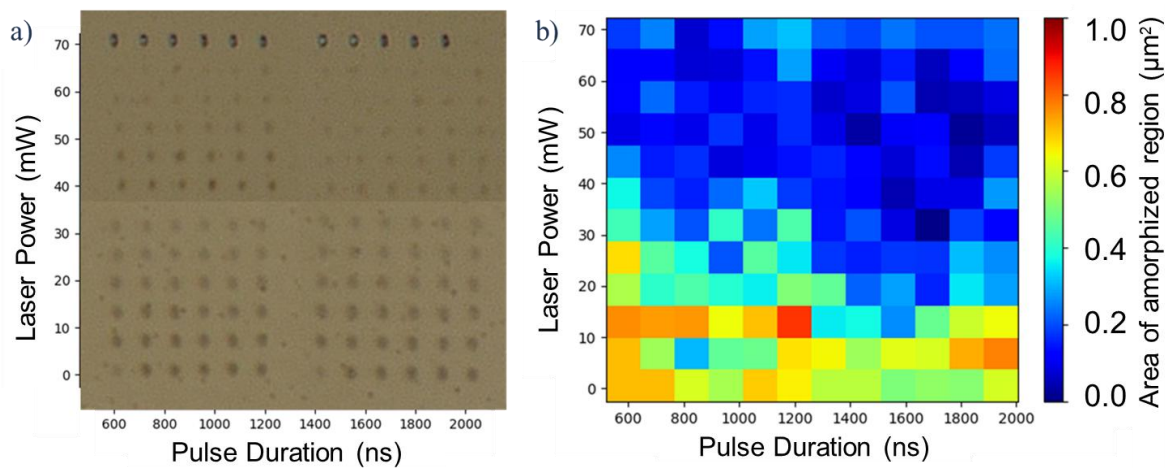


Figure 5.13: Recrystallization process of GeSb_2Te_4 . Response for different laser parameters, laser powers from 0 mW to 70 mW, pulse durations from 500 ns to 2000 ns and 20 repetitions, in the optical microscope a) and the corresponding PTE b).

To complement the optical and PTE observations, EBSD analysis was performed on the upper half of the switched region, where recrystallization is most likely to occur based on the applied laser parameters and optical microscope image. The EBSD scans are shown in Figure 5.14 as Inverse Pole Figure (IPF) maps overlaid with the Image Quality (IQ), which reflects the sharpness and contrast of the Kikuchi patterns and serves as a qualitative measure of crystallinity. The IQ contrast highlights differences in the quality of the diffraction patterns and makes it easier to identify features such as amorphous regions, grain

boundaries, or phase transitions, where the image quality typically drops due to structural disorder or changes in local symmetry. The EBSD maps reveal a variety of responses across different pulse conditions, allowing us to categorize the structural outcomes into distinct regimes.

At low pulse powers and short durations, the amorphous spots remain unchanged. These regions appear as black areas in the EBSD image, indicating the absence of Kikuchi patterns and confirming that the delivered energy was insufficient to initiate recrystallization. At the opposite end, in the row of the PTE diagram, corresponding to the strongest laser parameters, most of the switched spots appear severely disrupted or even destroyed. These regions either yield no EBSD signal or exhibit highly irregular patterns, suggesting that the high thermal load has led to ablation or irreversible material damage, rather than controlled recrystallization. These ablated or damaged spots are also clearly visible in the optical microscope, where they appear as dark spots, darker than the amorphous regions, with a lighter central area, likely corresponding to the physical void or “hole” resulting from ablation.

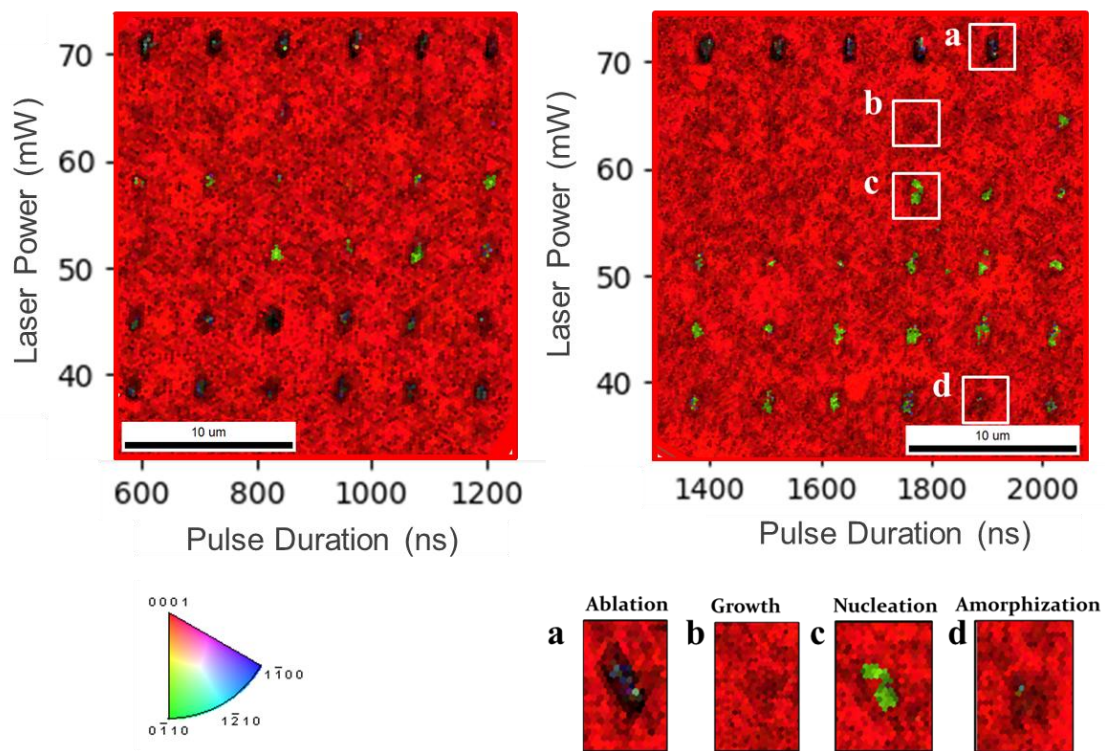


Figure 5.14: EBSD maps (IPF with IQ) corresponding to the upper half part of the PTE and four representative regions within the laser-switched area of the sample, illustrating the range of responses observed under varying pulse condition. **a)** Ablated region (high power, long pulse) showing no crystallographic information due to material damage. **b)** Growth-dominated recrystallization: the crystalline front propagates from the surrounding matrix, maintaining the original orientation. **c)** Nucleation-dominated recrystallization: new grains form within the amorphous region, often misoriented by $\sim 90^\circ$ relative to the surrounding matrix. **d)** Amorphous region (low power, short pulse) showing no EBSD signal, indicating the absence of recrystallization.

Between these two extremes lies a zone where successful recrystallization occurs, and two distinct mechanisms can be identified. At lower powers within this zone, nucleation-dominated recrystallization was observed, where new grains form within the amorphous region. These grains typically exhibit a 90° misorientation relative to the surrounding matrix. At higher powers within the recrystallization regime, growth-dominated recrystallization becomes the prevailing mechanism. In this case, the crystalline front propagates into the amorphous region from the surrounding matrix, preserving the original crystal orientation. The resulting structure retains the out-of-plane orientation of the as-deposited matrix, showing that the recrystallization front advances coherently from the edges of the amorphous region.

These observations are consistent with the known behavior of GST materials, where nucleation generally is favored at lower temperatures and growth becomes dominant as temperature or laser power increases. By identifying and distinguishing these recrystallization pathways in the EBSD data, we gain a deeper understanding of the structural dynamics governing reversible switching in phase-change materials.

N:2 and N:8 Superlattices

Following the reference study on GST-124, the recrystallization behavior of the superlattice (SL) samples was subsequently investigated, with particular focus placed on the two boundary cases: N:2 and N:8. These two SLs were selected to enable a comparative assessment across the structural modulation range. For each case, the optical microscope images of the switched regions were analyzed alongside the corresponding EBSD maps, allowing the degree and nature of recrystallization to be evaluated. (Figure 5.15) Based on these analyses, specific pulse conditions were identified that resulted in successful recrystallization, as indicated by optical re-brightening and the reappearance of crystalline orientation in the EBSD data.

In the superlattice samples, distinct recrystallization behaviors were observed depending on the layering scheme. For the N:2 SL, recrystallization appears to proceed exclusively via a growth-dominated mechanism, where the crystalline matrix surrounding the amorphous region drives the reformation of the structure, preserving its original orientation.

In contrast, the N:8 SL exhibits a recrystallization process, closely resembling that of the GST-124 reference sample. Here, both nucleation-dominated recrystallization, characterized by the formation of new grains with orientations differing from the original matrix, and growth-dominated recrystallization at higher laser powers were identified.

Based on these observations, suitable laser parameters for recrystallization were identified: 59 mW and 1735 ns for the N:2 SL, and 52 mW and 2005 ns for the N:8 SL. These values were selected from regions exhibiting clear growth-dominated recrystallization, ensuring

structural restoration aligned with the surrounding matrix, critical for reliable switching-cycle studies and subsequent APT analysis. Nonetheless, nucleation-dominated regions in the N:8 SL present an interesting opportunity for future investigations, as they could provide insight into the formation and evolution of independently recrystallized grains.

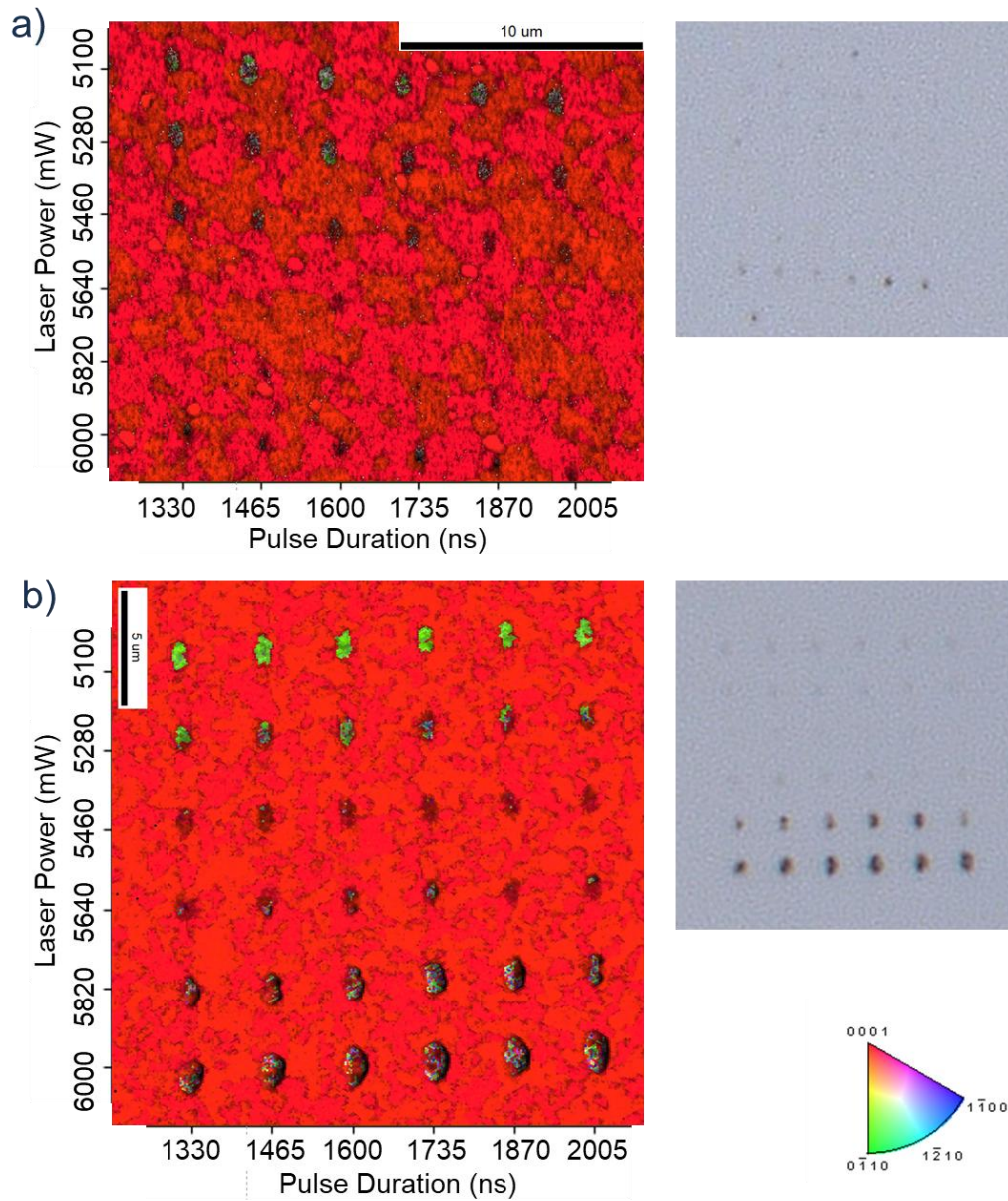


Figure 5.15 EBSD scans (IPF with IQ) for the a) N:2 and b) N:8 superlattices and accompanying optical microscopy images from the same area.

Recrystallized Superlattices in APT

To assess the state of the samples after recrystallization, APT measurements were performed on regions that had undergone recrystallization. This process followed the procedure outlined in the section “*Tip Preparation for Site-Specific Atom Probe Analysis of Recrystallized Regions*”. A key challenge encountered was achieving proper laser focus during the recrystallization process. In some cases, partial recrystallization occurred, as indicated by residual optical contrast in the previously amorphized regions. In this section, the APT results obtained from regions that exhibited no contrast with the surrounding crystalline matrix after recrystallization, considered indicative of successful structural recovery, are presented.

Additional liftouts were also taken from areas where recrystallization was partial. These results are provided in the APPENDIX, Figure 7.4 and Figure 7.5.

For the successfully recrystallized areas, in Figure 5.16, the reconstructed 3D atomic maps reveal that the layered structure is re-established, with the number of layers matching those observed in the as-deposited state. Specifically, for the N:2 superlattice approximately eight layers appear in the recrystallized structures, while in the N:8 configuration the four expected repetitions are visible. Additionally, the PME values increase again, reaching 62%, and 67% for N:2 and N:8 SLs, consistent with the restoration of a crystalline configuration and close to the as deposited PME values.

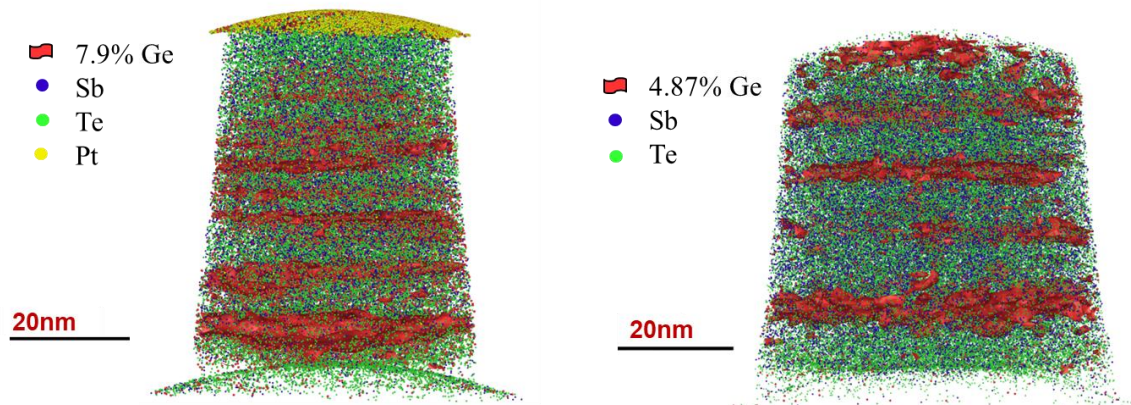


Figure 5.16: 3D atomic reconstructions of the recrystallized a) N:2 and b) N:8 superlattices, where for both SL configuration the layered structure is restored.

The corresponding elemental concentrations are listed in Table 4.

Strong intermixing remains evident, particularly for Ge, which shows a slight concentration increase in the Sb_2Te_3 layers after recrystallization. The resulting stoichiometries, especially in the GST-124 regions, still deviate considerably from the ideal ratios.

However, a notable shift toward the expected stoichiometry was observed across all layers. Ge content increased in the GST regions, suggesting a re-segregation of elements and

partial reformation of the intended layered structure. Similarly, Sb and Te concentrations in both the GST and Sb₂Te₃ layers moved closer to their nominal values. This trend points to the ability of the system to partially self-organize upon recrystallization, recovering some aspects of its designed composition and structure.

These observations confirm that recrystallization restores not only the crystalline phase but also the original superlattice architecture and promotes chemical reordering

Sample	Region	Ge (at.)	Sb (at.)	Te (at.)	Stoichiometry
N:2	GST- 124	7.9 ± 1.45	33.89 ± 0.84	57.92 ± 0.69	$\text{Ge}_{0.55 \pm 0.1} \text{Sb}_{2.34 \pm 0.06} \text{Te}_{4 \pm 0.05}$
	Sb ₂ Te ₃	3.74 ± 1.63	37.05 ± 1.36	59.9 ± 1	$\text{Ge}_{0.19 \pm 0.08} \text{Sb}_{1.86 \pm 0.07} \text{Te}_3 \pm 0.05$
N:8	GST- 124	4.65 ± 0.45	35.93 ± 0.37	58.93 ± 0.45	$\text{Ge}_{0.32 \pm 0.03} \text{Sb}_{2.44 \pm 0.03} \text{Te}_4 \pm 0.03$
	Sb ₂ Te ₃	2.34 ± 0.18	37.98 ± 0.28	60.51 ± 0.21	$\text{Ge}_{0.12 \pm 0.01} \text{Sb}_{1.88 \pm 0.01} \text{Te}_3 \pm 0.01$

Table 4: Elemental stoichiometries for the recrystallized N:2 and N:8 superlattices in the GST-124 and Sb₂Te₃ regions.

To further assess the multilayer structure after recrystallization, supercell thicknesses were evaluated from APT data and compared to the as-deposited state and from XRD values. (Figure 5.17 a) For the N:2 configuration, the average layer thickness in the recrystallized state was ~6.1% decreased compared to the as deposited. This close correspondence suggests that the original superlattice architecture is largely maintained upon recrystallization. In contrast, the N:8 configuration exhibited a more noticeable change, with the recrystallized average showing an ~8.9% increase relative to the as deposited superlattice. While both deviations are relatively small, the direction and magnitude of change may reflect subtle differences in recrystallization behavior between the two structures.

These trends are reflected in the PME comparison in (Figure 5.17 b) as well. While the N:2 recrystallized sample shows a relatively large error bar, the PME value remains comparable to that of the as-deposited state. For N:8, however, the PME is lower in the recrystallized state, suggesting potential changes in interfacial properties after structural recovery. It should however be noted that fewer successful APT tips were obtained for the recrystallized superlattices compared to the as-deposited state, which may introduce some variation in the derived values.

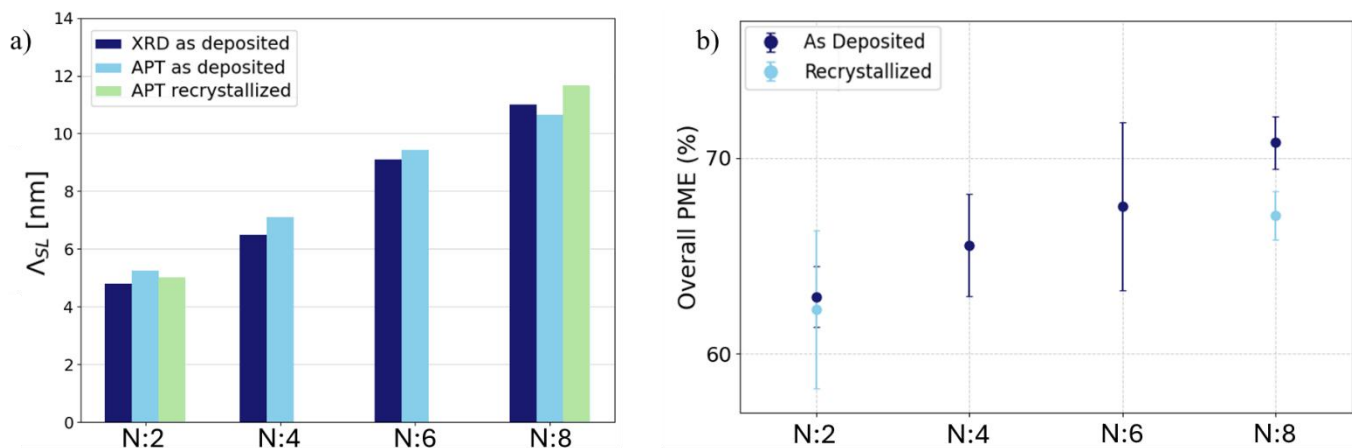


Figure 5.17 : a) Comparison of supercell thicknesses in the as-deposited and recrystallized states for N:2 and N:8 superlattices, based on APT and XRD data. b) PME values before and after recrystallization, highlighting consistent recovery for N:2 and a reduction for N:8

5.2.3 Stochastic behaviour in amorphous to crystalline transition

Nucleation is a stochastic process, which can be experimentally revealed by the variation in nucleation times observed under similar conditions. To better understand how this stochasticity occurs during the amorphous-to-crystalline phase transition, crystallization measurements were carried out using a constant laser pulse power of 52 mW and varying pulse duration 0 – 2300 ns in the LSS setup for both the N:2 and N:8 samples.

In Figure 5.18 for both samples, a transition boundary was observed, marking a switch from the amorphous to the crystalline phase. Around this boundary, a spread of responses was found, some points remained amorphous, others were fully crystalline, and several showed intermediate values. This distribution of outcomes under identical pulse conditions confirms the stochastic nature of the transition.

The amorphized state is represented by the light blue markers, where each is induced by an identical laser pulse of 170 mW and 16 ns, prior to the application of any recrystallization pulses. The resulting amorphized spots are also showing measurable variation despite being created under the exact same conditions.

Notably, the recrystallized state, characterized by no clearly observable spot, is reached faster in the N:2 sample, at approximately 1400 ns, compared to 1900 ns for the N:8 sample. This indicates a shorter recrystallization time for N:2. However, it is also evident that the spread of measurement points during the transition region is broader for the N:2 sample

than for N:8, suggesting greater variability and more pronounced stochasticity in the phase change for the N:2 configuration.

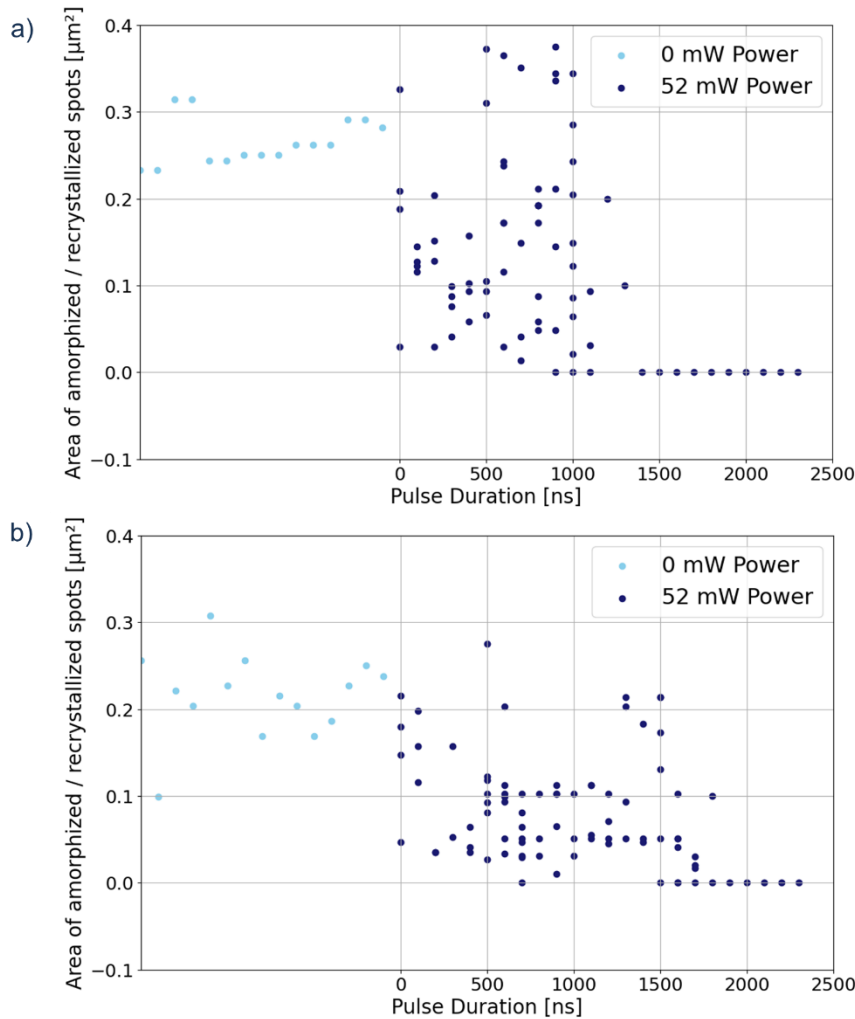


Figure 5.18: a) N:2 and b) N:8 SL. Laser-induced response as a function of pulse duration. Dark blue markers show the measured spot area at various pulse durations from 0 to 2300 ns. Light blue markers represent baseline measurements taken at 0 mW power (before applying recrystallization pulses). The data illustrate a transition from the amorphized state to laser-induced recrystallization with increasing pulse duration.

5.3 Investigation of bond confinement effects in GeSb_2Te_4 / Sb_2Te_3 Superlattices

The investigation into bond confinement effects began with an observation: among the as deposited superlattices, the N:2 configuration consistently exhibited significantly lower probability of multiple events (PME) values compared to the other samples. PME, which reflects the likelihood of detecting multiple ions on the same laser pulse during Atom Probe Tomography (APT), is sensitive to the nature of atomic bonding. This prompted the hypothesis that reduced thickness of the Sb_2Te_3 layers in the superlattices could lead to increased covalency in these structures. This change in bonding character, introduced by bond confinement effects, can in turn influence the switching mechanism. Specifically, we propose that thinner Sb_2Te_3 layers, which have been shown in previous studies to show

bond confinement effects for decreased thin film thickness [56, 57], ultimately lead to improved switching performance. This observation motivated a broader investigation across all superlattice configurations.

To test this hypothesis, two key parameters, PME and amorphized spot size, were analyzed across the four superlattice configurations and GeSb_2Te_4 , as indicators of the bonding mechanism and switching performance, respectively.

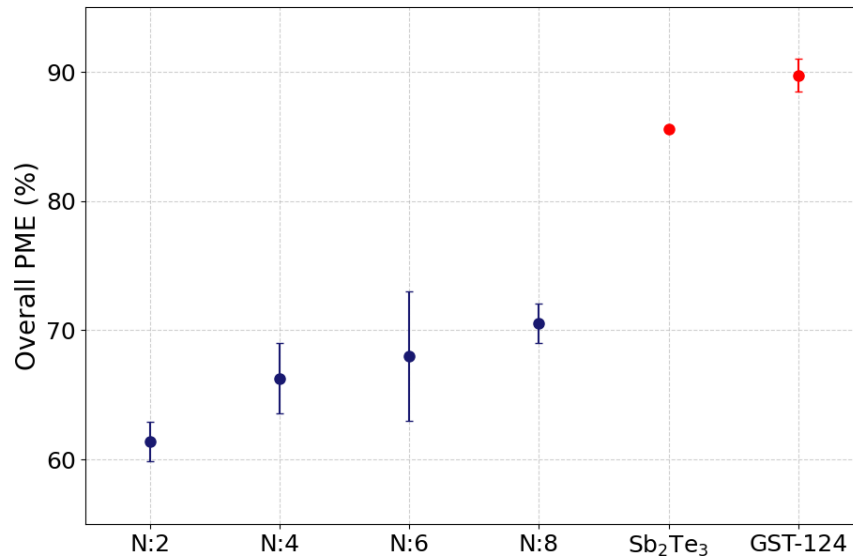


Figure 5.19: Overall PME values for the different samples. Superlattices show lower PME (~60–70%) compared to GST-124 (90%) and Sb_2Te_3 (86%). A trend of decreasing PME with thinner supercells is also observed, with N:2 showing the lowest values.

The PME values in this case represent averages over at least three measurements per sample, considering only successful measurements, defined as those showing at least half of the expected superlattice layers. From the PME analysis (Figure 5.19), it is evident that the $\text{Ge}_1\text{Sb}_2\text{Te}_4/\text{Sb}_2\text{Te}_3$ superlattices exhibit PME values in the 60–70% range, substantially lower than the ~90% measured for GeSb_2Te_4 . The GST-124 reference, with PME value of above 80%, aligns with earlier results for compounds along the GeTe – Sb_2Te_3 pseudobinary line. Notably, a trend emerges within the superlattice series: thinner supercells (with N:2 as the thinnest) correspond to lower PME values. PME reflects the degree of metavalent bonding, characterized by delocalized p-electrons, while lower PME indicates increased localization and covalent character. This shift toward more covalent bonding, associated with the PME reduction is described as bond confinement. Varying the Sb_2Te_3 thickness tunes bond confinement strength: thicker layers retain more metavalent character, while ultrathin films enhance localization and covalency. This is reflected in the decreasing PME with Sb_2Te_3 layer thickness across the superlattice series, highlighting layered structure configuration as a key factor in the bonding interplay. Despite the relatively large error bars, particularly for the N:6 sample, the trend remains consistent. Interestingly, the N:6 sample systematically deviated from expected behavior in the optical switching measurements and

appears to be an outlier, despite no obvious structural irregularities or fabrication-related factors. Additionally, to better understand the bonding behavior in the superlattices, the PME signal was also analyzed separately for GST-like and Sb_2Te_3 -like regions using APT data in the APPENDIX, Figure 7.7.

To quantify switching efficiency across the samples, the laser power required to produce a given amorphized spot size, measured by optical microscopy, was analyzed. (Figure 5.20) A set of representative spot areas between 0.2 and $0.9 \mu\text{m}^2$ was selected, each corresponding to reliably amorphized regions, which exclude destroyed areas. For these areas, the GST-124 reference consistently required the highest power, ranging from 200 to 240 mW, with larger spot sizes demanding more energy. In contrast, the superlattices exhibit a general downward trend in required power as the Sb_2Te_3 layer thickness decreases. Although not strictly monotonic for every spot size, the overall trend is clear: superlattices with thinner supercell thickness, particularly the N:2 configuration, require significantly less power, ranging between 120 and 180 mW. This reduction in power is most pronounced at $0.3 \mu\text{m}^2$, which also corresponds to the amorphized spot size previously selected for further experiments. These results clearly demonstrate the enhanced switching performance of superlattices with thinner supercells and confirm that the amorphized spot size can serve as a reliable indicator of switching efficiency.

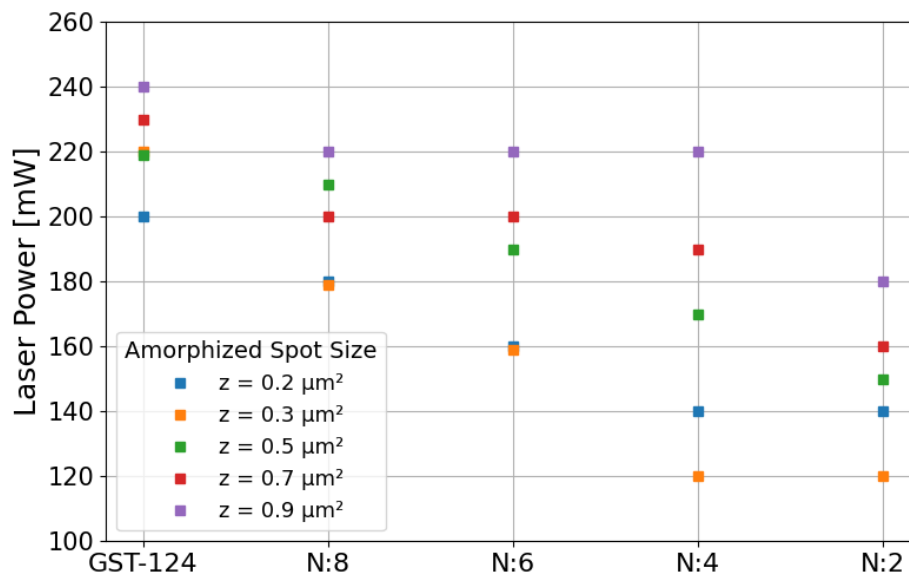


Figure 5.20: Laser power required to induce various amorphized spot sizes (0.2 – $0.9 \mu\text{m}^2$). GST-124 shows the highest switching power, while superlattices, especially the thinner N:2 configuration, require significantly less, indicating enhanced switching efficiency.

Notably, the spread in required laser power for a given range of amorphized spot sizes is larger for the superlattices than for GST-124. This broader distribution may reflect the increased tunability of the switching behavior in superlattices and is consistent with the wider range of resistance states observed during electrical switching. [7]

When the overall PME is plotted against the amorphized spot size, a clear correlation emerges, lower PME values are associated with smaller amorphized areas. This trend supports the hypothesis that bonding confinement effects in thinner superlattices enable more efficient and controlled switching.

However, the exact mechanism linking increased covalency, or bond confinement, to improved switching performance remains unclear. While the observed correlation between lower PME and reduced switching power is compelling, further investigations are needed to confirm a causal relationship. Specifically, additional characterization techniques will be required to probe how changes in bonding character influence material properties such as electrical resistivity, thermal conductivity, or optical contrast in superlattices. Only once these intermediate steps are clarified can a comprehensive hypothesis be formulated, describing how increased covalency in Sb_2Te_3 at the atomic level results in macroscopic improvements in switching efficiency.

To begin addressing this gap, our ongoing work includes phonon measurements using a femtosecond (fs) pump–probe setup and thermal conductivity measurements using frequency-domain thermoreflectance (FDTR), intended also at comparing heat confinement effects under electrical and optical switching conditions. These techniques are specifically aimed at revealing trends and differences among the different superlattice configurations, as well as in comparison to the GeSb_2Te_4 reference and pure Sb_2Te_3 . By tracking systematic variations in phonon lifetimes, thermal conductivity, and energy dissipation pathways, we aim to determine whether superlattices with thinner supercells, such as N:2, exhibit altered lattice dynamics or enhanced thermal confinement due to changes in bonding character. Preliminary results from phonon and thermal conductivity measurements are presented in the APPENDIX, Figure 7.8 and Figure 7.9.

CONCLUSION AND OUTLOOK

This work aimed to investigate the structural and switching behavior of $\text{GeSb}_2\text{Te}_4/\text{Sb}_2\text{Te}_3$ superlattices (SLs) and explored how their nanoscale configuration influences their performance as phase-change materials. A comprehensive picture of the structural evolution and switching mechanisms across SLs of varying supercell thickness has been built through a combination of atom probe tomography (APT), laser switching experiments, EBSD analysis, and optical microscopy characterization.

Structural Characterization and APT Insights

Despite the nanometric scale of the superlattice layers, APT enabled detailed analysis of the as-deposited SLs, revealing off-stoichiometry, intermixing, and defect-related features. These observations confirm that meaningful structural information can be extracted even from ultrathin SLs approaching the dimensions relevant to real-world device architectures. This pushes the limits of APT application in phase-change materials and demonstrates its viability as a tool for investigating and resolving sub-5 nm engineered materials, as seen for the N:2 SL.

Switching Mechanism, Amorphization Evidence and Opportunities

One central question addressed in this study was the nature of the switching mechanism in SLs. Two competing hypotheses exist in the literature: a melt-quench process versus an order-to-order transition. Our results strongly support the melt-quench mechanism, primarily due to the loss of the layered superlattice structure in the amorphous state, as revealed by APT. Furthermore, in the recrystallized regions, the layered architecture was restored, confirming the reversible phase-change behavior of these SLs and the robustness of their engineered structure. Acquiring amorphous-state APT data was challenging, since there is a possibility of recrystallization by the pulsed laser during analysis, and successful amorphous data sets were limited. Nevertheless, a particularly valuable addition would be

APT measurements of amorphized regions in the N:2 SL, which could examine whether the PME trend continues in the amorphous state and shed light on how much of the SL character persists post-switching.

Recrystallization Behavior and Future Directions

The loss of the original layered structure in the amorphous state and the subsequent recovery of it in recrystallized superlattices was one of the most striking results. However, several aspects of the recrystallization process continue to raise open questions. In common GST, nucleation and growth pathways followed well-established behavior. In the SLs, particularly N:2 and N:8, APT measurements of the recrystallized areas were focused on growth-dominated regions, yet EBSD revealed nucleation-dominated areas as well, characterized by grain orientations distinct from the original crystalline matrix. These merit further exploration, APT studies of such regions could offer important insights into alternative recrystallization pathways in SLs. Additionally, varying the size of the recrystallized laser spot in future experiments could help better isolate and compare nucleation- and growth-dominated regimes, offering more precise spatial control and deeper understanding of the recrystallization dynamics.

From Interface Hypothesis to Bond Confinement Hypothesis

Initially, based on the consensus that the vdW-like gap structures within the material layers play a crucial role [58], the switching performance of PCM superlattices was hypothesized to be enhanced primarily through interface effects. However, this perspective was revised in light of the observed trends in PME values. Improved switching in superlattices with thinner supercells was found to coincide with lower PME values, indicative of increased covalency. As a result, a bond-confinement-driven mechanism was proposed, in which the reduced Sb_2Te_3 thickness is believed to enhance bond confinement, thereby influencing the switching dynamics and enabling more energy-efficient transitions. To further explore the bond confinement hypothesis and its impact on switching behavior, several complementary techniques are proposed:

- **TEM**, to reveal atomic-scale structural details, elemental distributions, and bonding environments across the different SL configurations.
- **Phonon measurements**, to probe vibrational confinement effects and reveal trends among the SLs.
- **Thermal conductivity measurements**, to assess heat transport and evaluate the degree of thermal confinement and its potential contribution to improved switching efficiency.
- **Density Functional Theory (DFT) calculations** can model bond lengths, bonding character, and thermal properties, linking atomic structure to observed switching trends.

Toward Deeper Understanding and Device Relevance

Taking all the above into account, these findings contribute to building a more complete model of the switching mechanism in superlattices. By linking structural configuration to band confinement and improved phase-change behavior, this work demonstrates that superlattices as thin as 60 nm can achieve significant performance gains. This brings them closer to realistic device dimensions and lays the groundwork for establishing design principles for next-generation phase-change memory devices.

APPENDIX

Superlattice configuration scheme

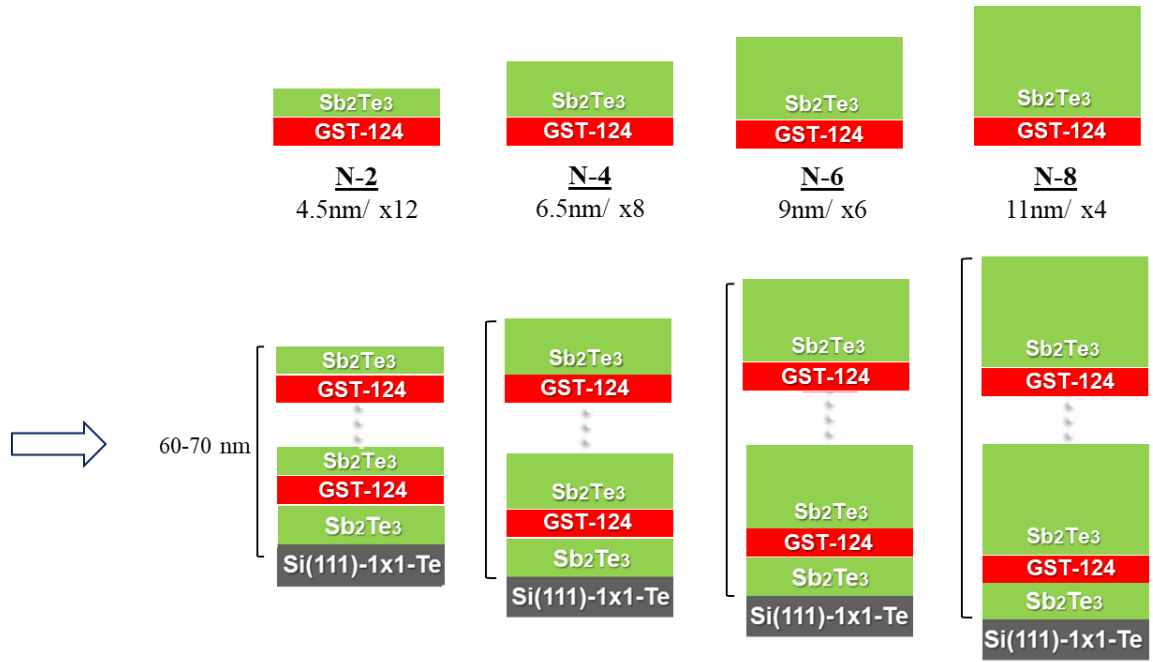


Figure 7.1: Schematics of the N-X series of GST-124 / Sb₂Te₃ superlattices grown by molecular beam epitaxy (MBE). The top row shows the supercell structure of each configuration (N-2, N-4, N-6, N-8), consisting of two GST-124 septuple layers and an increasing number of Sb₂Te₃ quintuple layers from 2 to 8 per supercell. These variations were designed to investigate the effect of layer architecture on superlattice properties. The lower row illustrates the full stack of each sample, starting from a Te-passivated Si(111)-1×1 substrate, followed by a base Sb₂Te₃ layer. The supercells are then repeated until a total film thickness of approximately 60–70 nm is reached, with Sb₂Te₃ also used as the final layer.

GST-124 reference in APT

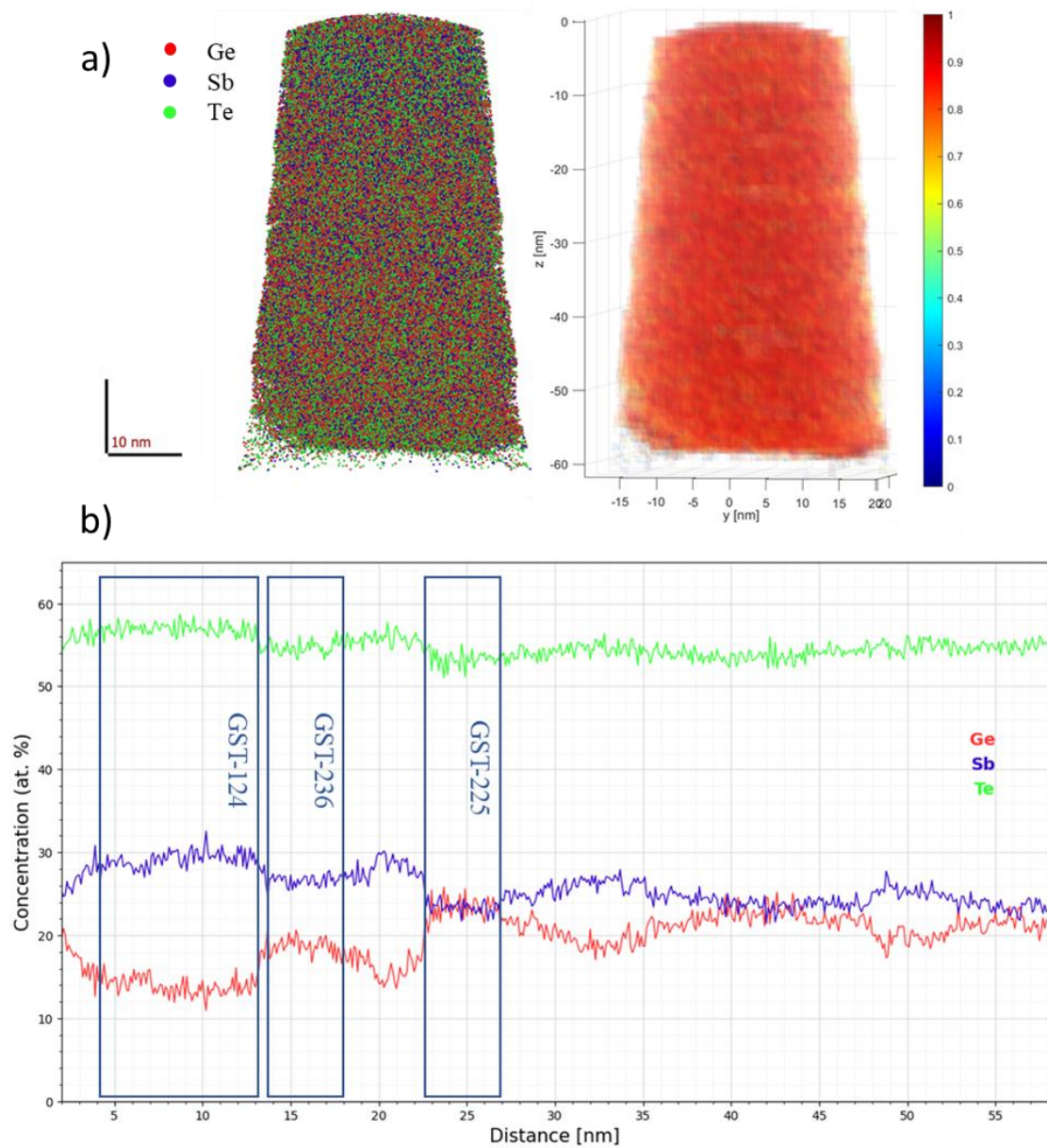


Figure 7.2: (a) 3D elemental map and 3D multiplicity of the GST-124 reference sample, showing a multiplicity of $\sim 89\%$. (b) Corresponding concentration profile reveals the presence of multiple GST configurations in the tip, beyond the targeted GST-124.

Additional EBSD results

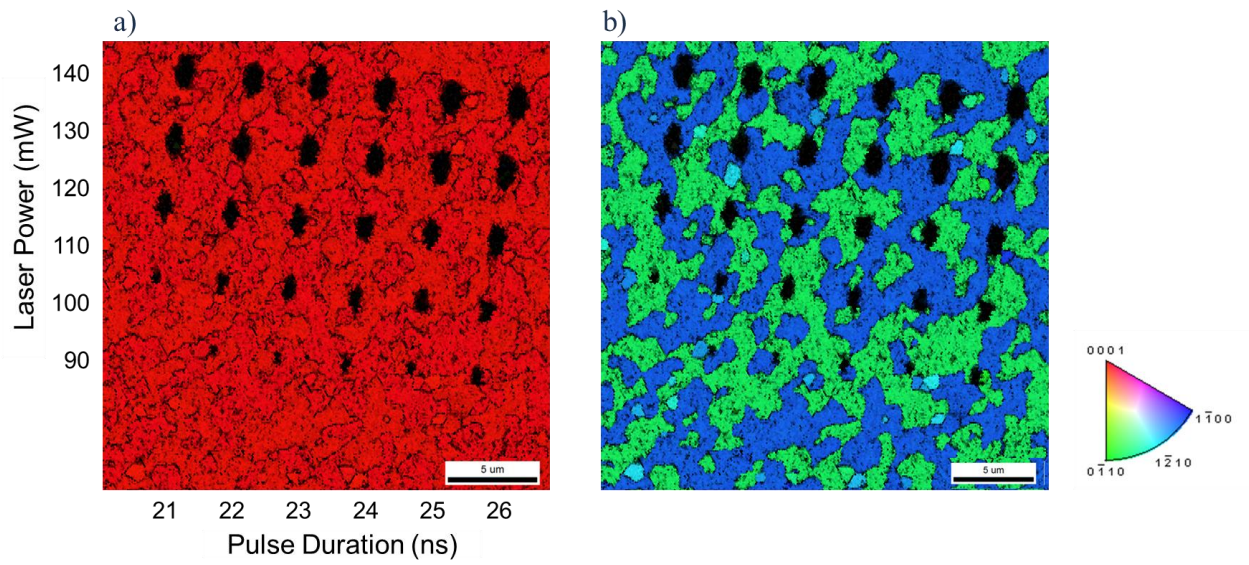


Figure 7.3: EBSD images of the Inverse Pole Figure (IPF) overlaid with the confidence index (CI) for both out-of-plane (a) and in-plane (b) orientations of the N:2 superlattice (SL). A strong out-of-plane texture is observed, while the in-plane orientation exhibits typical twinning along with the presence of side domains. Amorphized regions, induced by laser parameters in the range of 90–140 mW and 21–26 ns, appear as black areas. The threshold for amorphization can be identified, with the first amorphized spots observed at 100 mW and 22 ns. Twinning and side domains are prominent in the in-plane orientation.

Partially Recrystallized Areas measured in APT

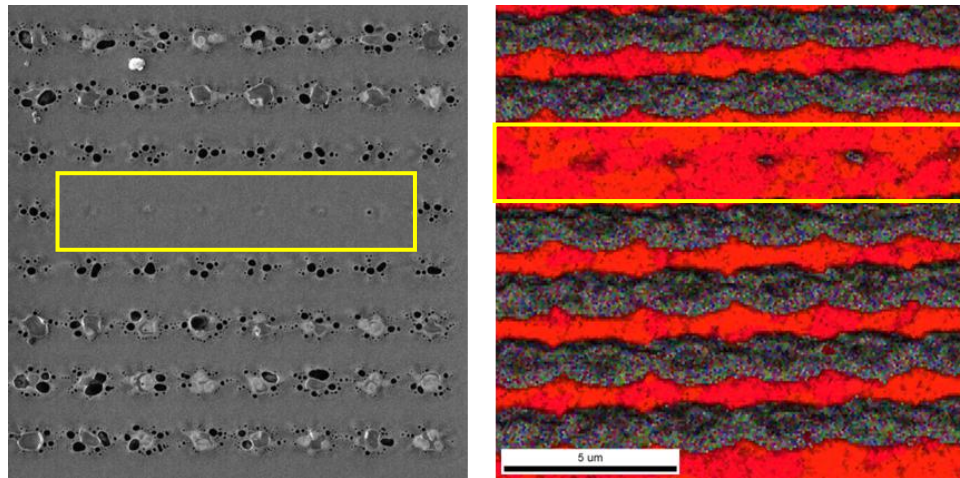


Figure 7.4: (Left) SEM image showing residual structures (highlighted in the yellow box) resulting from out-of-focus recrystallization pulses, surrounded by damaged areas created according to the process described in the section “Tip Preparation for Site-Specific Atom Probe Analysis of Recrystallized Regions.” (Right) EBSD scan (Inverse Pole Figure overlaid with confidence index) of the same region reveals areas with contrast relative to the red crystalline matrix, indicating partial recrystallization. These regions are smaller than the original amorphized spot and are also surrounded by the destroyed spots.

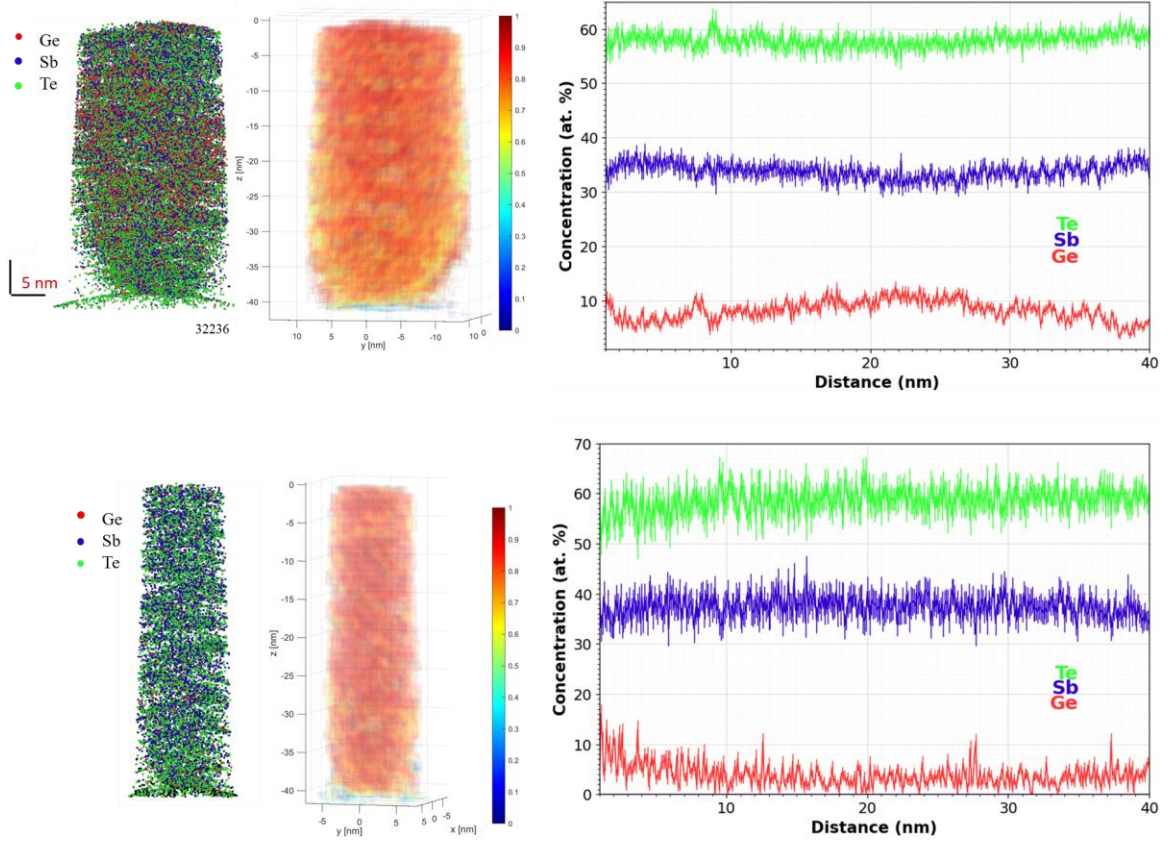


Figure 7.5: APT analysis of superlattice regions where recrystallization pulses were applied with suboptimal focus, resulting in partial recrystallization. **(Top):** 3D reconstruction, PME distribution, and concentration profile for the N:2 sample. The concentration profile reveals a lack of periodic modulation in elemental distribution, indicating no recovery of the original layered structure. The overall PME is at ~80%, significantly above previous PME values (~65%) for the N:2 superlattice. **(Bottom):** A similar case for the N:8 sample showing no superlattice structure and a PME of ~85%, higher than in the as deposited crystalline superlattice. Both the concentration profiles and the PME values indicate a state closer to that of the reference crystalline GST-124 or Sb_2Te_3 than to superlattices.

Optical Switching parameters

Sample	Initial Amorphization [x 1 Pulse]		Recrystallization [x 20 Pulses]		Reamorphization [x 1 Pulse]	
	Laser Power [mW]	Pulse length [ns]	Laser Power [mW]	Pulse length [ns]	Laser Power [mW] (step 10 mW)	Pulse length [ns] (step 1 ns)
N:2	170	16	58.8	1735	90 – 260	9 – 26
N:4	170	16	45.1	1600	90 – 260	9 – 26
N:6	170	16	52	1870	90 – 260	9 – 26
N:8	170	16	52	2005	90 – 260	9 – 26
GST- 124	170	16	58.8	1735	90 – 260	9 – 26

Table 5: Laser parameters used during an optical switching cycle: amorphization – recrystallization - reamorphization. The process consists of an initial single pulse amorphization step, followed by a set of 20 identical recrystallization pulses, and a range of reamorphization single pulses to evaluate the switching performance of the samples. All steps are conducted using the same repetition rate.

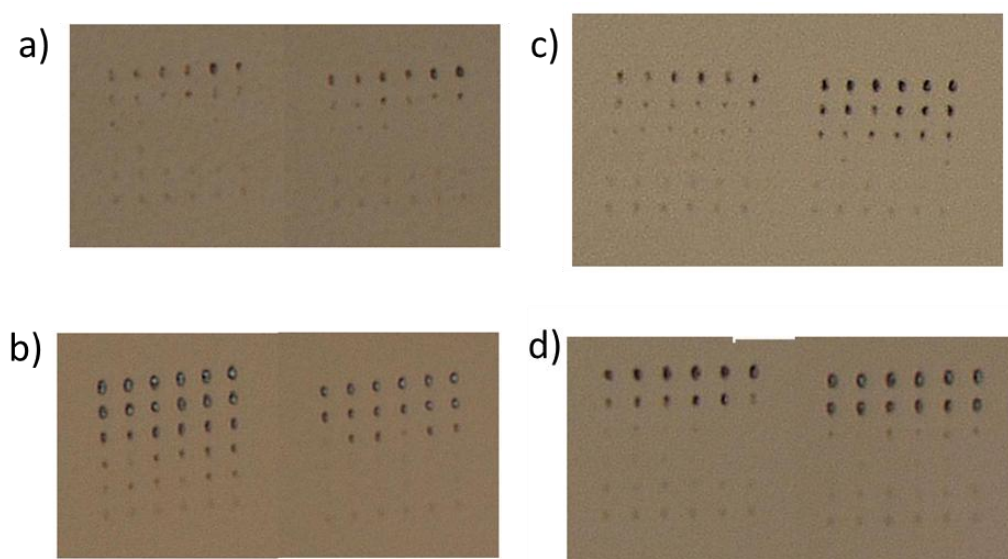


Figure 7.6: Optical microscope images showing the recrystallization of superlattices (a) N:2, (b) N:4, (c) N:6, and (d) N:8. Recrystallization pulses with power from 38 to 72 mW (bottom to top in each superlattice) and pulse duration from 520 to 2005 ns (left to right) were applied to determine the optimal laser parameters for recrystallization for each superlattice, these optimal parameters are summarized in Table 5.

Investigation of PME in individual layers

To better understand the bonding behavior in the superlattices, the PME signal was analyzed separately for GST-like and Sb_2Te_3 -like regions using APT data. Ge-containing ions were used to represent GST-like layers, while all other species were attributed to Sb_2Te_3 . Despite some intermixing at the interfaces, this classification provides a useful first-order approximation.

PME values from the GST-like regions showed a reduction, typically 3–6%, compared to the total average, suggesting slightly stronger covalent character potentially influenced by confinement. In contrast, PME from the Sb_2Te_3 layers closely matched the overall values, within 0.5%, indicating they dominate the bonding character.

3D PME distribution maps (Figure 7.7) restricted to Ge-containing ions revealed periodic variations aligned with the superlattice structure, supporting the interpretation that bonding modulation is spatially resolved and not random.

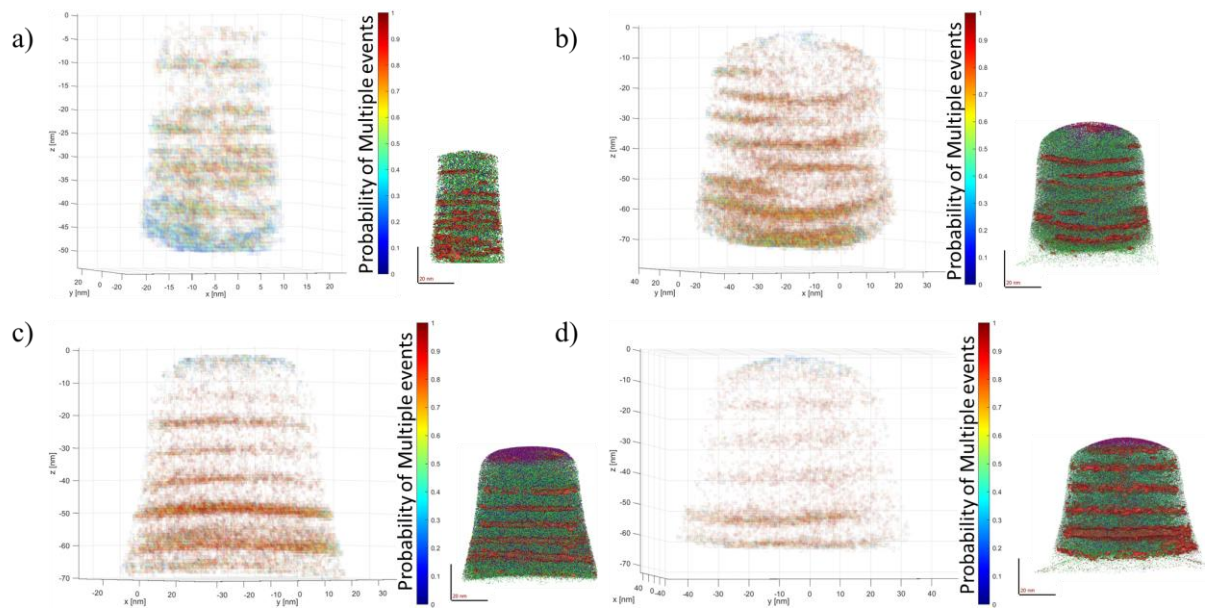


Figure 7.7: 3D PME distribution selected only for ions that include Ge for a) N:2, b) N:4, c) N:6 and d) N:8 SL. Accompanying 3D atomic reconstruction.

Preliminary Results from Phonon and Thermal Conductivity Measurements

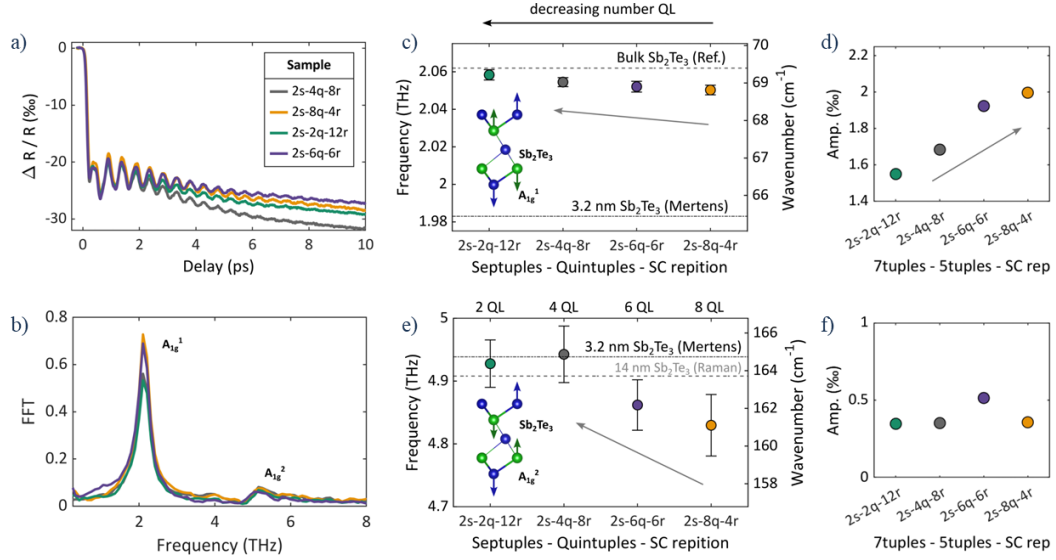


Figure 7.8: Anisotropic optical fs pump-probe spectroscopy results for superlattice samples with varying period thicknesses. (a) Normalized transient reflectivity ($\Delta R/R$) as a function of pump-probe delay time. (b) Corresponding FFT spectra showing the observed coherent phonon modes A_{1g}^1 and A_{1g}^2 . (c–d) Extracted frequency and amplitude of the A_{1g}^1 mode across samples. A trend toward higher phonon frequencies was observed with decreasing supercell thickness, or with reduced Sb_2Te_3 layer thickness, opposite to the behavior reported in [56]. (e–f) Frequency and amplitude of the A_{1g}^2 mode.

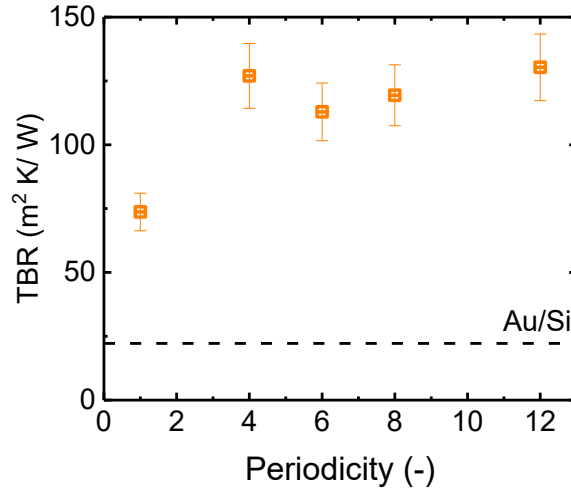


Figure 7.9: Thermal boundary resistance (TBR) as a function of superlattice periodicity, measured using frequency-domain thermoreflectance (FDTR). Each data point corresponds to a different number of supercell repetitions, with periodicities of 4, 6, 8, and 12 representing superlattices N: 8, N:6, N:4, and N:2, respectively. Increase of the TBR, results in a decrease in the thermal conductivity of the PCM, thereby confining the heat within the material. This enhances the thermal efficiency of the material, and consequently, the energy required for phase transition is reduced compared to common GST.

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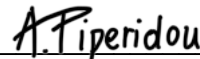
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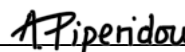
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